ECOCAR INNOVATION CHALLENGE · PRODUCT INNOVATION TRACK · CHECK-IN 1 (B-1.3.1 TO B-1.3.3) · STUDIO 1 (B-2.3.1 TO B-2.3.4)

Western Michigan University

Where an engineering team can win in the software-defined vehicle

Our Studio 1 business portfolio: how value moves in the software-defined vehicle, how incumbents make money, and three wedges we will test with real engineers in Studio 2.

9 actor groups	**8** value-chain layers
43 claims audited	**3** wedges for customer discovery

How it is organized: five chapters that follow the Studio 1 brief, then an appendix. See the contents and the Check-In map.

Updated 11 Nov 2026 · market research 13 Oct 2026 · company screen 26 Oct 2026

EXECUTIVE SUMMARY

The opening is the work every OEM must finish before launch

Software-defined vehicles move a car's value to a few central computers and their software, and let automakers keep earning after the sale. The biggest pools belong to well-funded incumbents, so our opening is the work every OEM must finish before launch: integration, verification and validation (V&V). It is the fastest-growing slice of auto software spend, and it gates every release.

$519B
automotive software & electronics by 2035, 4.5%/yr vs ~1% for vehicles [1]

11.4%/yr
integration, verification & validation services: the fastest-growing slice of auto software spend [1]

77%
of E/E architectures expected to be zonal or central by 2035, up from under 30% [1]

~15%
of recalled US vehicles since 2022 were eligible for an over-the-air fix [3]

Three wedges for customer discovery

W1 · Safety evidence for ML components
First to test · From the company screen: 20 of 25 with dSPACE · DFV 64% · confidence low

W2 · What can leave the bench
Weakest today · From the company screen: 17 of 25 with dSPACE, conditional · DFV 57% · confidence low

W3 · Model-update re-test scoping
Newest · Added after the screen; closest fit to our GM CAV work · DFV 61% · confidence low

Our first screen pointed to HIL, zonal and OTA. Testing them against real companies dropped zonal and OTA and narrowed HIL. See how our thinking changed.

First claim we test for each wedge

1. W1: Safety engineers have had evidence for an ML component challenged in ways sampled testing cannot answer.

2. W2: HIL leads have kept or delayed tests because no measured data showed what a virtual ECU would miss.
3. W3: Teams that retrain perception models re-test far more than a change needs, or cut scope on judgment they struggle to defend.

CONTENTS

Five chapters, in the order of the Studio 1 brief

> **1. The market**
> B-2.3.1
> How is value moving in the software-defined vehicle, and who captures it?
> Why SDVs · Ecosystem · Value chain stages · Value chain · Where value grows

> **2. How incumbents make money**
> B-2.3.2
> How do companies at different stages of the chain capture value today?
> Business models

> **3. Where we fit**
> B-2.3.3
> Where could our skills give us an edge, and who are the engineers involved?
> Round 1 shortlist · Company screen · How our thinking changed · Engineering personas

> **4. Our wedges**
> B-2.3.4
> What exactly will we test, and what would make us drop it?
> Wedge hypotheses

> **5. Next steps**
> How we reach real engineers in Studio 2, and what we still need
> Outreach plan · Innovation in society · Risks & gaps · Questions for you

Check-In 1: B-1.3.1 Value chain analysis and business model canvases (chapters 1–2); B-1.3.2 Wedge hypotheses and rationale, with the weakest named (chapters 3–4); B-1.3.3 Outreach pipeline, matched to our personas (chapter 5). Appendix: Round 1 scoring · Dossiers · Evidence audit.

DESIRABILITY

The problem, who has it, and what would be better
Wedge hypotheses (Overview tab) Executive summary

Ecosystem diagram and value chain map
Ecosystem Value chain

Specific personas, and each wedge tied to a persona's pain
Engineering personas Wedge hypotheses ("Who has it" tab)

FEASIBILITY

The weakest technical assumption behind each wedge
Wedge hypotheses (Market & assumptions tab)

Engineering pain points on the value chain, and where our skills fit
Value chain (click any layer)

VIABILITY

Sourced practice canvases at different stages of the chain
Business models

Early market logic, and the missing pieces
Wedge hypotheses (Market & assumptions tab)

INNOVATION PROCESS QUALITY

Why these wedges, which is weakest, and our gaps with an action each
How our thinking changed Wedge hypotheses (DFV scoring) Risks & gaps

Next tests, desirability first, with evidence sources
Wedge hypotheses ("Full test plan" tab)

An outreach pipeline matched to the personas
Outreach plan Engineering personas

INNOVATION IN SOCIETY

Safety, security, privacy, fairness and environment
Innovation in society

Sourced claims, labelled assumptions, respectful outreach, no OEM data
Evidence audit Outreach plan (outreach rules)

COMMUNICATION

Can a reviewer follow the logic without us?
Executive summary How it is organized

Our 30-minute walkthrough
Questions for you

CHAPTER 1 · THE MARKET · WHY SDVS

Why companies are betting on SDVs

Automotive software and electronics: $519B by 2035, growing 4.5% a year against ~1% for vehicles [1].

Recurring revenue

$3.1B

GM guides ~$3.1B of OnStar revenue for 2026 [6] and targets ~70% gross margins on OnStar and Super Cruise [8].

Cheaper hardware

17 → 7

Rivian's Gen 2 R1 cut ECUs from 17 to 7 [11].

Cheaper recalls

29.3M

vehicles in 891 US recalls in 2025 [2]. OTA fixes could save US OEMs $1.5B by 2028 (2023 forecast) [4].

Faster development

18–24 mo

Chinese rivals target 18–24-month cycles, about half the Western norm [12].

CHAPTER 1 · THE MARKET · ECOSYSTEM

The OEM is still the hub, but SDVs add a second revenue loop after the sale

Automakers (OEMs): the hub

Brings

Vehicles, features and subscriptions; consented data to partners.

Gets paid

Before SDVs: Vehicle margin, paid once at the sale. With SDVs: Vehicle margin plus recurring software revenue.

Risk

The OEM is still the hub, but SDVs add a second revenue loop after the sale. The bet is risky: VW's Cariad lost €2.4B in 2023 and again in 2024 [37], and Ford scrapped its FNV4 architecture in 2025 [38].

Regulators & standards

NHTSA · UNECE R155/R156 · US Commerce (BIS) · FTC · ISO 26262 / 21434

Rules, type approval and recalls. UN R155/R156 has been mandatory for all new EU registrations since July 2024 [26]; the BIS rule bars Chinese/Russian connected-vehicle software from MY2027 and hardware from MY2030 [30]; the FTC's Jan 2026 order makes opt-in consent the gate for data revenue [10]. ISO 26262 and ISO/SAE 21434 are voluntary standards that approvals lean on.

Supply side

Chipmakers

SoCs, MCUs, SDKs

Brings

SoCs, MCUs and SDKs, sold to Tier 1s and, increasingly, straight to OEMs.

Gets paid

Chip price per unit plus multi-year platform deals.

What SDVs change

OEMs increasingly buy central compute straight from chipmakers: BMW and Qualcomm co-developed Snapdragon Ride Pilot [15] and signed a long-term compute supply deal in 2026 [16].

Tier 1 suppliers

controllers, modules, integration

Bosch · APTIV · ZF

Brings

Controllers and modules, plus the integration work that puts them into a vehicle.

Gets paid

Module price per vehicle.

What SDVs change

As OEMs buy compute directly, Tier 1s lean toward integration work (the shift is not yet quantified; chipmakers still partner with Bosch and others). Harman agreed to buy ZF's ADAS business in Dec 2025 [42]. Zonal ECU migration, our Round 1 target #2, is part of this integration work; we dropped it after the company screen.

Software platforms

OS, hypervisor, middleware

QNX · VECTOR > · Android

Brings

The certified operating system, hypervisor and middleware everything else runs on.

Gets paid

Per-unit royalty that runs for the vehicle's whole production life.

What SDVs change

Merchant vendors are concentrating (QNX, Google, Vector, Applied Intuition), OEMs build their own (RV Tech) [20], and OEM-backed open source pressures price [19].

SDV specialists

our focus area

OTA, cybersecurity, V&V tools

PlaxidityX DRIVING TRUST · dSPACE · excelfore

Brings

OTA platforms, cybersecurity and V&V tooling: the jobs every OEM must finish before launch.

Gets paid

Licences and SaaS subscriptions.

What SDVs change

Our focus area sits here: integration, verification and validation (V&V), and specifically whether a model, a test or an update can be trusted. All three wedges are V&V work, with dSPACE and Applied Intuition as the routes in. OTA and SBOM evidence were dropped.

Cloud & connectivity

backend, cellular, compute

aws · Azure · verizon

Brings

Cloud backend, cellular connectivity and compute.

Gets paid

Usage fees.

What SDVs change

Every connected feature adds backend and cellular load; data volume vs cellular cost is a standing pain point.

Demand side

Dealers & service

sales, delivery, repair

franchise dealers · *repair shops*

Brings

Sales, delivery and repair to consumers.

Gets paid

Buy vehicles at wholesale; earn on sales and service.

What SDVs change

Dealers lose software-fix visits as updates move over the air, though in ABI's analysis nearly half of software-recall vehicles still needed a dealer visit [4].

Consumers

keep paying for services after the sale

OnStar · Super Cruise

Brings

Usage data back to the OEM, and monthly payments for features and subscriptions such as OnStar and Super Cruise (GM services shown as examples).

Gets paid

Features and subscriptions delivered after the sale.

What SDVs change

Drivers keep paying after the sale: GM guides ~$3.1B of OnStar revenue for 2026 [6]. About 30–40% of Super Cruise owners keep paying after the 3 included years [7].

Fleet operators

rental, ride-hail and robotaxi fleets

Hertz · Uber · Waymo

Brings

Fleet data back to the OEM, and fleet contracts.

Gets paid

Vehicles and APIs.

What SDVs change

Fleets buy on uptime and total cost of ownership, and often run third-party telematics beside the OEM's own.

Data partners

telematics, maps, location data

Samsara · tomtom · HERE

Brings

Telematics, map and location products built on vehicle data.

Gets paid

Consented vehicle data, under a data licence.

What SDVs change

The FTC's final order against GM (14 Jan 2026) makes opt-in consent the gate for data revenue [10].

What SDVs change

1. **Chips go direct.** OEMs increasingly buy central compute straight from chipmakers (BMW–Qualcomm, 2025–26) [16], pushing Tier 1s toward integration work.
2. **Paying after sale.** Drivers and fleets keep paying the OEM after the sale. GM guides ~$3.1B of OnStar revenue for 2026 [6] and targets ~70% gross margins on OnStar and Super Cruise [8].
3. **Fixes go OTA.** Dealers lose software-fix visits as updates move over the air. About 15% of recalled US vehicles since 2022 were eligible for an OTA remedy [3].
4. **3rd-party telematics.** Fleets buy on uptime and total cost of ownership, and often run third-party telematics beside the OEM's own.

Before SDVs: One revenue loop: suppliers sell through Tier 1s, the OEM sells the car once through dealers, and a software fix means a dealer visit.

Flows

From → to	Product or service	Paid back	Data	SDV shift
Chipmakers → Automakers	design wins	platform deals	—	1 new with SDVs
Tier 1 suppliers → Automakers	controllers	module price	—	
Software platforms → Automakers	OS, middleware	royalty	—	

From → to	Product or service	Paid back	Data	SDV shift
SDV specialists → Automakers	OTA, cyber, V&V	licences, SaaS	—	
Cloud & connectivity → Automakers	backend, cellular	usage fees	—	
Chipmakers → Tier 1 suppliers	SoCs	chip price	—	
Automakers → Dealers & service	vehicles	wholesale	—	3
Automakers → Consumers	features, subs	monthly	usage data	2 new with SDVs
Automakers → Fleet operators	vehicles, APIs	fleet contracts	fleet data	
Automakers → Data partners	consented data	data licence	data	new with SDVs
Dealers & service → Consumers	sales, service	—	—	
Data partners → Fleet operators	3rd-party telematics	—	—	4 new with SDVs

CHAPTER 1 · THE MARKET · VALUE CHAIN STAGES

All three wedges sit inside the EcoCAR discovery zone

The organizers' seven-stage value chain, filled in with our research.

	1. Compute & sensing OEM has it covered · our layers L01, L03	2. E/E architecture OEM has it covered · our layers L01	3. Vehicle OS & middleware Adjacent · our layers L02, L04, L07	4. Control & autonomy stack Discovery zone · our layers L03	5. Data, simulation & validation Discovery zone · our layers L08	6. In-cabin AI & HMI Discovery zone	7. Monetization & lifecycle OEM has it covered · our layers L05, L06
Does	Turns the road into data and runs it: sensors, chips and central computers.	Wires computers, power and controllers into zones around a central backbone.	The car's operating system: installs, isolates, secures and updates all software.	Perceives, predicts and plans the drive, from driver assistance to automated driving.	Proves the vehicle works: simulation, HIL benches, virtual ECUs and test toolchains.	Makes automation understandable and trustworthy for the person in the seat.	Keeps the car earning after the sale: subscriptions, fleet services and data products.
Handoff	Raw sensor signals and the compute headroom every later stage depends on.	Reliable, on-time data transfer to the controls.	A certified platform that lets software above it ship once, on many chips.	Turns trusted sensor data into safe motion.	Turns requirements into evidence that regulators and buyers accept.	Turns machine confidence into human trust.	Turns everything upstream into recurring revenue.

	1. Compute & sensing OEM has it covered · our layers L01, L03	2. E/E architecture OEM has it covered · our layers L01	3. Vehicle OS & middleware Adjacent · our layers L02, L04, L07	4. Control & autonomy stack Discovery zone · our layers L03	5. Data, simulation & validation Discovery zone · our layers L08	6. In-cabin AI & HMI Discovery zone	7. Monetization & lifecycle OEM has it covered · our layers L05, L06
Example challenge	Hardware is frozen years before the software it must run exists.	Splitting, porting and re-testing old controller software over years of mixed fleets.	Bridging old AUTOSAR Classic controllers with Linux-based platforms, and patching for a 15-year vehicle life.	Proving safety across rare edge cases, especially for learned (ML) components.	HIL benches are scarce, and virtual ECUs miss timing and I/O detail.	Handing control back to a driver who has stopped paying attention.	Every OEM exposes different signals, and consent now gates data revenue [10].
Stable	A chip design win locks in for the platform's whole life.	Zonal-plus-central layouts are where architectures are heading [1].	Certified platforms (QNX, Android Automotive, Vector); cybersecurity rules are mandatory for EU sales [26].	Hands-free highway driver assistance is deployed at scale (Super Cruise, Mobileye, Tesla).	Model-, software- and hardware-in-the-loop test pipelines are standard practice.	Infotainment platforms and driver-monitoring cameras ship today.	Paid connected services (OnStar) and fleet telematics (Samsara, Motive).
In motion	OEMs now buy compute directly from chipmakers.	Moving functions off legacy controllers, model year by model year.	Large OEMs bringing OTA in-house; the US connected-vehicle software rule [30].	New safety guidance for AI components (ISO/PAS 8800); automated-driving timelines keep slipping.	The fastest-growing slice of auto software spend [1]; testing ML-based features.	Conversational AI assistants in the car.	Factory-fit telematics replacing aftermarket boxes; stricter consent for driving data.
Where we sit	Not a target: chipmakers such as Qualcomm, NVIDIA and Mobileye dominate.	Zonal ECU migration. The work lives inside OEM programs.	OTA for commercial vehicles, and SBOM & cyber evidence.	**W1 W3** W1 and W3 start here: evidence for a learned perception component.	**W2 W1 W3** W2 sits here, and W1 and W3 deliver their evidence here.	Not mapped in depth in Studio 1. Infotainment apps sat in our commodity quadrant.	Not taken forward: mixed-make fleet data; edge-side data selection only as an extension of W1.

Format and zones from the organizers' example; content is our research. The positions we dropped (zonal migration, OTA, SBOM) all sat outside the discovery zone.

Six layers from silicon to paid services, and two that run through all of them

Competition ratings and open pockets are team hypotheses. W1 and W2 mark where two of our wedges sit (W3 sits with W1 in the ADAS layer); dropped marks earlier targets.

06 Data & cloud services

Vehicle data platforms, cloud backend and data products

CONTESTED | EVERY MONTH | Edge-side data selection

20 years of opt-in consent required by the FTC's final order against GM, 14 Jan 2026 [10]

aws | Azure | Google Cloud | tomtom | HERE | *OEM data platforms*

WHAT WINNING LOOKS LIKE

- Turn consented data into predictive maintenance, insurance and ADAS-training value
- Players: hyperscalers, OEM data platforms, fleet software firms

KPIs: revenue per connected vehicle, consent rate, data cost per vehicle

MOATS & GROWTH DRIVERS

- Installed base plus consent
- The FTC order bars GM from sharing driving data with consumer reporting agencies for 5 years [10]
- GM has accrued $0.5B for related investigations and litigation [9]

CRITICAL, AND WHERE COMPETITION IS LOW

- Privacy-safe consent now gates monetization
- Direct resale of driver data looks unattractive

Edge-side selection of useful data

ENGINEERING PAIN POINTS

- Data volume vs cellular cost
- Inconsistent signal schemas across OEMs
- Consent tracking; data quality

Where our skills fit: Partial. Cloud and data skills; edge-side data selection only as an extension of W1.

05 Telematics & connectivity

Connected services, fleet telematics and cellular

CROWDED | EVERY MONTH | Mixed-make fleet data

$3.1B OnStar revenue GM guides for 2026 (+15%), from ~12M subscribers at ~$20 average monthly revenue each [6]

OnStar | motive | Samsara | Geotab | AT&T | verizon

WHAT WINNING LOOKS LIKE

- A large connected base that pays every month
- Samsara: $2.1B ARR, +30% YoY (quarter to Aug 2026) [23]

KPIs: ARR, net revenue retention, ARPU, trial-to-paid conversion

MOATS & GROWTH DRIVERS

- Installed base and data scale (Samsara: 25T+ data points a year) [24]
- Workflow lock-in for fleets
- Growth: AI safety features; factory-fit telematics replacing aftermarket boxes

CRITICAL, AND WHERE COMPETITION IS LOW

- Access to raw vehicle data from the OEM
- Aftermarket fleet telematics is crowded

Normalizing data across mixed-make fleets

ENGINEERING PAIN POINTS

- Each OEM exposes different signals and APIs
- Heavy-duty (J1939) and light-duty vehicle buses differ
- Cellular cost; signal latency and gaps

Where our skills fit: Partial. CAN and J1939 skills, but on their own they would not stand out in a crowded telematics market.

04 OTA & software lifecycle

Over-the-air updates, rollback and campaign management

CONTESTED | PER VEHICLE + UPGRADES | ★ OTA for commercial & off-highway DROPPED

~15% of recalled US vehicles since 2022 were eligible for an over-the-air remedy [3]; 891 recalls covered 29.3M vehicles in 2025 [2]

TESLA | RIVIAN | HARMAN | excelfore | Sibros | Elektrobit

WHAT WINNING LOOKS LIKE

- Full-vehicle updates (every ECU, not just infotainment) with near-zero failed installs
- Software-first OEMs such as Tesla and Rivian treat OTA as a core in-house capability; specialists such as Harman, Sibros and Excelfore supply many others [41]

KPIs: share of ECUs updatable OTA, success rate, campaign duration, recalls fixed OTA

MOATS & GROWTH DRIVERS

- Deep bootloader and ECU integration makes vendors sticky
- UN R156 software-update compliance [26]
- Growth: recall pressure; OTA fixes could save US OEMs $1.5B by 2028 (ABI forecast, 2023) [4]

CRITICAL, AND WHERE COMPETITION IS LOW

- Safe rollback and version dependencies across ECUs
- Large OEMs increasingly insource

Full-vehicle OTA for commercial & off-highway

ENGINEERING PAIN POINTS

- Legacy ECUs without dual (A/B) memory
- Long flash times take vehicles out of service; cellular data cost
- Cross-ECU compatibility testing; most recalls are still hardware, though software is now ~20–25% of campaigns [40]

Where our skills fit: Weak. Embedded and cloud skills, but no PKI, secure-boot or UN R156 experience. Dropped after the company screen.

03 ADAS & automated driving

Perception, planning and driver assistance, L2 to L4

CROWDED | PER VEHICLE + UPGRADES | ★ Safety evidence for ML W1

$9B → $50B ADAS/AD software, 2025–35: the largest software growth pool, nearly 20%/yr (McKinsey: 19.4%) [1]

mobileye | TESLA | Super Cruise | NVIDIA | Qualcomm | WAYMO

WHAT WINNING LOOKS LIKE

- High attach rates at a low per-unit price, plus paid upgrades
- Mobileye: ~38M EyeQ units in its 2026 outlook; Surround ADAS at a $100–150 ASP and ~70% gross margin [21]
- Tesla: 1.48M active FSD (Supervised) users in Q2 2026, about 45% on subscription [22]

KPIs: ASP, attach and renewal rate, disengagements, MTBF

MOATS & GROWTH DRIVERS

- Fleet-data flywheels
- Accumulated safety-validation evidence
- Growth: L2 could be 52% of sales by 2030, L3 16% by 2035 [1]

CRITICAL, AND WHERE COMPETITION IS LOW

- Proving safety across the long tail of edge cases
- Very high competition in L2+ software stacks

★ W1 · Safety evidence for ML components

ENGINEERING PAIN POINTS

- Sensor calibration and fusion
- Scenario coverage for validation
- Compute and LiDAR cost per vehicle; L3/L4 timelines keep slipping

Where our skills fit: Strong for W1 and W3: CAV, ADAS and perception skills, simulation, and verification research through our faculty advisor. Not a fit for building an L2+ driving stack.

02 Software platforms

Operating system, hypervisor and middleware

CROWDED | ONCE PER VEHICLE | Consolidating: little open space

275M+ vehicles running QNX, across 45+ automakers including the top 10 [17]

QNX | Android Automotive | VECTOR > | WNDRVR | Elektrobit (Aumovio) | RV Tech (Rivian–VW)

WHAT WINNING LOOKS LIKE

- Be the certified layer everything else runs on
- Android Automotive powers Volvo/Polestar, GM, Renault and others; Ford's system is built with Google [18]

KPIs: vehicles deployed, royalty backlog, OEM programs won, safety certifications

MOATS & GROWTH DRIVERS

- Safety certification is slow and costly to copy
- Per-unit royalties run for the vehicle's whole production life
- Growth: OS and middleware ~$3B → ~$13B, 2025–35 (16.2%/yr) [1]

CRITICAL, AND WHERE COMPETITION IS LOW

- Hardware abstraction that lets apps move across chips and model years
- Merchant vendors are concentrating (QNX, Google, Vector, Applied Intuition), beside OEM-built platforms such as Rivian–VW's RV Tech [20]
- OEM-backed open source (Eclipse S-CORE, still pre-1.0) is likely to pressure middleware pricing, so there is little open space here [19]

ENGINEERING PAIN POINTS

- Bridging legacy AUTOSAR Classic ECUs with Linux and Adaptive platforms
- Mixed-criticality hypervisors; porting across SoCs
- Maintaining software over a 15-year vehicle life

Where our skills fit: Weak. Certified operating systems and middleware are beyond a student team; we use these platforms rather than build them.

01 Compute & E/E

Central, zonal and domain controllers; SoCs and MCUs

CROWDED · ONCE PER VEHICLE · ★ Zonal ECU migration DROPPED

$5B → $17B central compute units, 2030–35; zonal units grow $4B → $13B [1]

Qualcomm · NVIDIA · mobileye · NXP · Infineon · RENESAS · Bosch · APTIV

WHAT WINNING LOOKS LIKE

- Multi-year design wins carried across many OEM platforms
- Qualcomm: record $1.59B automotive revenue in fiscal Q3 2026, +61% YoY [13]; Snapdragon Cockpit in 75M+ vehicles (June 2025) [14]

KPIs: design-win pipeline, content per vehicle, TOPS per watt, ASIL rating, win-to-SOP time

MOATS & GROWTH DRIVERS

- A design win locks in for the platform's whole life
- SDK ecosystems raise switching cost
- OEMs now buy compute directly: BMW and Qualcomm co-developed Snapdragon Ride Pilot [15]

CRITICAL, AND WHERE COMPETITION IS LOW

- Sizing compute headroom: analysts advise 30–50% spare for future OTA features, more in premium cars [5]
- Central SoCs are intensely contested

Zonal I/O and power design; legacy ECU migration

ENGINEERING PAIN POINTS

- Hardware is frozen years before the software it must run exists
- Thermal and power budgets; memory supply constraints
- Partitioning safety-critical and non-critical workloads on one chip

Where our skills fit: Partial. Embedded and CAN skills plus our test vehicle help with zonal I/O work, but migration needs AUTOSAR and ISO 26262 depth we lack. Dropped after the company screen.

08 Integration, V&V & tooling

Simulation, hardware-in-the-loop (HIL) test, virtual ECUs and development toolchains

CONTESTED | SPANS EVERY LAYER | ★ What can leave the bench W2

11.4%/yr integration, verification and validation services: the fastest-growing slice of auto software spend, 2025–35 [1]

Applied Intuition | dSPACE | VECTOR | SYNOPSYS | *OEM in-house HIL labs*

WHAT WINNING LOOKS LIKE

- Own the test loop that OEMs run every day
- Applied Intuition: $15B valuation (June 2025) [32], with ~$30B offers reported Oct 2026 [33]; says it serves 18 of the top 20 OEMs; ~$830M ARR (2025 estimate) [34]

KPIs: ARR, OEM coverage, test-cycle time, rig utilization

MOATS & GROWTH DRIVERS

- Embedded in OEM CI pipelines and safety evidence, so hard to rip out
- 52% of engineering VPs name integration complexity as a top SDV challenge [36]
- Synopsys now includes Ansys simulation (deal closed July 2025) [43]

CRITICAL, AND WHERE COMPETITION IS LOW

- In QNX's 2025 survey, developers cite long cycles (37%) and debugging and testing (36%) [35]
- The tool market is fragmented below the leaders

★ W2 · What can leave the bench (SIL vs HIL)

ENGINEERING PAIN POINTS

- HIL benches are expensive and scarce
- Virtual ECUs lack timing and I/O fidelity
- Embedded CI is hard because code depends on hardware; ISO 26262/21434 traceability

Where our skills fit: Strong for W2: we run HIL benches and have controls and model-based design skills. Against dSPACE this gives us empathy, not an advantage.

07 Cybersecurity

The gate to type approval: no compliant cybersecurity management system, no sales

CONTESTED · SPANS EVERY LAYER · ★ SBOM & supply-chain evidence · SCREENED OUT

494 public automotive cyber incidents in 2025; 92% remote, 44% ransom-related (Upstream) [25]

Upstream · VicOne (Trend Micro) · PlaxidityX DRIVING TRUST · Karamba · *OEM vehicle SOCs*

WHAT WINNING LOOKS LIKE

- Pass type approval and run a vehicle security operations centre (vSOC) at fleet scale
- PlaxidityX (formerly Argus Cyber Security) is now part of Aumovio, Continental's spin-off [44]

KPIs: vehicles monitored, time to detect and respond, audit pass rate

MOATS & GROWTH DRIVERS

- Regulation is the engine: UN R155/R156 mandatory for all new EU registrations since July 2024 [26]
- China (GB 44495, in force Jan 2026) [27] and Korea (CSMS, Aug 2025) [28] enforce aligned rules; India's AIS-189/190 are still draft [29]
- US Commerce (BIS) bars Chinese/Russian connected-vehicle software from MY2027 and hardware from MY2030 [30]

CRITICAL, AND WHERE COMPETITION IS LOW

- No compliant cybersecurity management system, no sales
- Moderate competition among specialists

★ **SBOM and supply-chain provenance evidence**

ENGINEERING PAIN POINTS

- Patching over a 15-year vehicle life; key management
- SBOMs across many suppliers
- Small ECUs that can't run heavy crypto; 42 ISO/SAE 21434 work products [31]

Where our skills fit: Weak. Our cybersecurity background is coursework, and audit-grade evidence needs credibility we lack.

CHAPTER 1 · THE MARKET · WHERE VALUE GROWS

Where the money grows

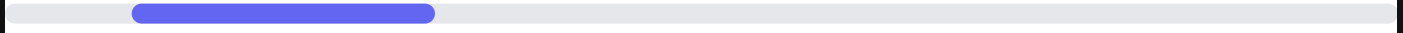
The biggest pools are per-vehicle software and compute, in crowded layers; recurring services are smaller but growing.

ADAS / AD software
2025–35 · crowded



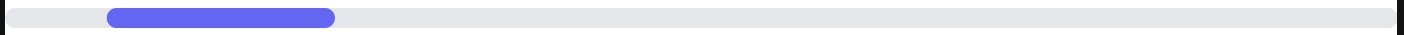
$9B → **$50B** · 19.4%/yr

Central compute units
2030–35 · crowded



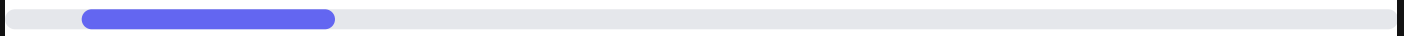
$5B → **$17B** · 27.7%/yr (computed)

Zonal units
2030–35 · crowded



$4B → **$13B** · 26.6%/yr (computed)

OS & middleware
2025–35 · crowded



~$3B → **~$13B** · 16.2%/yr

Every pool ends in 2035, so end values compare like with like; start years differ (2025 or 2030). Source: McKinsey [1]. All four sit in layers we rate crowded; our wedges are evidence work beside them, not rivals in them. McKinsey gives no comparable pool for telematics, data or OTA.

Every month: telematics, data & cloud

Smaller, but still growing: GM guides OnStar to ~$3.1B in 2026, +15% [6]; Samsara's ARR is up 30% a year [23].

Integration and V&V: 11.4%/yr

The fastest-growing slice of auto software spend, in McKinsey's cut by activity rather than domain [1]. All three of our wedges are validation work.

CHAPTER 2 · INCUMBENTS · BUSINESS MODELS

How three incumbents make money

Three companies at different stages of the value chain, and what each one taught us about where a student team could fit.

dSPACE

20 of 25: lead passes

Test and simulation tools that help carmakers validate vehicle software before it reaches the road [51]

>€460M revenue 2024, +14% · **>2,900** employees · **1988** founded, private

HOW IT MAKES MONEY

- Test hardware systems (HIL, prototyping) (core)
- Software tool licences (core)
- Engineering services, maintenance and training (core)
- Cloud validation (SIMPHERA) (growth)

WHAT PROTECTS IT

Benches, models and test cases built on dSPACE get reused for years, so switching is costly (analyst view).

Privately held, so revenue and staff figures are company-reported rather than from audited filings.

WHAT IT TAUGHT US

It owns HIL, so we should not compete there. Its public materials describe simulation checks for its neural-network workflow, and it sponsors the challenge.

Full canvas (PDF, 33 cited sources)

Applied Intuition

19 of 25: passes if we find a contact

Vehicle software, simulation and autonomy for cars, trucks, mining, construction, agriculture and defense [50]

$15B valuation, Jun 2025 · **~$400M** ARR end-2024, reported · **18 of 20** top non-Chinese carmakers

HOW IT MAKES MONEY

- Software licences by seat, workload and module (reported) (core)
- Vehicle OS and autonomy licensing (terms undisclosed) (growth)
- Integration and customization services (reported) (core)
- Government contracts, including a CDAO ceiling of up to $171M (growth)

WHAT PROTECTS IT

Lands with one tool, then expands. Engineers embed with customer teams on multi-year partnerships.

ARR is press-reported, not company-disclosed. Contract ceilings are not revenue.

WHAT IT TAUGHT US

Its evidence is statistical and scenario-based, while its safety roles ask for ISO/PAS 8800 and UL 4600. No warm way in yet.

Full canvas (PDF, 35 cited sources)

Mobileye

16 of 25: near-miss on access

Vision chips and driving software sold per vehicle; robotaxis and robots are early bets [52]

$1.89B revenue FY2025, +15% · **~$48.5** average per system, Q2 2026 · **~4,130** employees; Intel owns ~77%

HOW IT MAKES MONEY

- EyeQ chip sales through Tier 1s: about 92% of revenue (core)
- SuperVision and premium ADAS systems, per vehicle (growth)
- Cloud-Enhanced ADAS, with higher profit per unit (growth)
- Robotaxi rides (target ~$125k per vehicle a year) (future bet)

WHAT PROTECTS IT

Repeat awards from all top-10 customers, locked in at design win, with 250M+ vehicles built with EyeQ.

The $3.8B goodwill writedown in Q1 2026 is non-cash: it drives the GAAP loss but is not an operating cost.

WHAT IT TAUGHT US

A very different model: hardware margin per car, sold through Tier 1s. Strong safety-methods culture, but no route in for us yet.

Full canvas (PDF, 25 cited sources)

The brief asks for two companies at different stages of the value chain; we built three. Data as of 30 Oct 2026. One point is only half confirmed: the dSPACE canvas lists NVIDIA as a partner for AI scene generation. dSPACE confirms the NuRec integration, but no primary source places it in NVIDIA Halos.

CHAPTER 3 · WHERE WE FIT · ROUND 1 SHORTLIST

Round 1 pointed to four whitespace pockets

Placed by criticality to launch and competitive intensity: team assessments, not measurements.

Engineering whitespace

critical, fragmented

- **2** Zonal ECU migration
- **1** HIL & virtual-ECU validation
- **3** OTA for commercial vehicles
- ✕ SBOM & cyber evidence

Crowded core

critical, crowded

- Central SoCs
- L2+ ADAS stacks
- Vehicle OS

Unproven

less critical, fragmented

- Edge-side data selection
- Driver-data resale

Commodity

less critical, crowded

- Aftermarket fleet telematics
- Infotainment apps

1. HIL and virtual-ECU validation

Round 1 target #1

52% of engineering VPs name integration complexity as a top SDV challenge [36], and HIL benches are costly and scarce.

Personas: V&V leads, test-bench engineers

Narrowed → W2: dSPACE already owns HIL tools, so we kept one question: what can leave the bench. 17 of 25, a conditional pass.

2. Zonal migration of legacy ECUs

Round 1 target #2

McKinsey expects 77% of architectures to be zonal or central by 2035, up from under 30% today [1], so legacy ECU software must be split, ported and re-validated over years of mixed fleets.

Personas: E/E architects, systems integration engineers

Dropped: The work lives inside OEM programs and needs AUTOSAR and ISO 26262 depth we lack. Best company score 13 of 25.

3. OTA for commercial & off-highway

Round 1 target #3

Large OEMs insource OTA; smaller and specialty makers still need full-vehicle updates on mixed hardware.

Personas: embedded leads, fleet maintenance managers

Dropped: Large OEMs insource it, Applied Intuition bundles it, and open-source tools exist. Best company score 11 of 25.

✕ SBOM & cyber lifecycle evidence

screened out in Round 1

UN R155 type approval [26] and the US Commerce rule's compliance declarations [30] turn supplier provenance and patching into gates.

Personas: product cybersecurity engineers, supplier quality

Dropped twice: Buyers need audit credibility a student team lacks, and specialists already compete (team fit 2.8 of 5). None of the three companies sells SBOM tools: best score 10 of 25.

Why we dropped SBOM & cyber evidence

The need is real: cybersecurity rules gate EU type approval [26], and the US connected-vehicle rule needs supplier declarations [30].

1. **Round 1 · 5 Oct: Team fit too low.** Feasibility 2.8 of 5, #5 of 7. Buyers need audit credibility we lack, and specialists already compete.
2. **Round 2 · 26 Oct: No company pull.** None of the three companies sells SBOM tooling, so none would pull our work. Best score 10 of 25.

Cybersecurity stays a requirement of whatever we build. Reopens only with a cybersecurity faculty partner and a supplier partner.

CHAPTER 3 · WHERE WE FIT · COMPANY SCREEN

Tested against real companies, 3 of 15 pairs pass

Can we do the work, can we reach them, and what can we add that they lack?

Position	dSPACE	Applied Intuition	Mobileye
W1 · Safety evidence for ML Re-scoped from ADAS validation evidence	**20** of 25 Pass	**19** of 25 Pass, access-dependent	**16** of 25 Near-miss: fails on access
W2 · What can leave the bench SIL-vs-HIL fidelity, narrowed from HIL & virtual-ECU validation	**17** of 25 Conditional pass, deferred	**15** of 25 Fail: total	**10** of 25 Fail
Zonal I/O & migration Round 1 target #2	**13** of 25 Fail	**12** of 25 Fail	**7** of 25 Fail
OTA, commercial & specialty Round 1 target #3	**11** of 25 Fail	**9** of 25 Fail	**7** of 25 Fail

Position	dSPACE	Applied Intuition	Mobileye
SBOM & provenance Already screened out in Round 1	**10** of 25 Fail	**10** of 25 Fail	**9** of 25 Fail

A pair passes only if Expertise fit and Accessibility are both at least 3 and the total is at least 17 of 25. Differentiation of 2 or less makes it a conditional pass, pursued only alongside a full pass. Mixed-make fleet data and edge-side data selection were not scored against the companies. The report's verdicts: fleet data is not reconsidered for now; edge-side selection only as an extension of W1. [49]

All 15 pairs, and what decides each

Pair	Fit	Diff	Need	Access	Strat	Total	What decides it
W1 · Safety evidence for ML × dSPACE	5	4	3	4	4	**20**	Its public materials describe simulation checks for its neural-network workflow, and dSPACE sponsors the challenge. Three points of margin. Access falls to 3 unless licence scope is confirmed by 15 Dec 2026.
W1 · Safety evidence for ML × Applied Intuition	4	4	3	3	5	**19**	Safety roles ask for ML-safety skills, but there is no EcoCAR link. A named contact must agree to talk by 15 Dec 2026. Two points of margin, but Access sits on the floor of 3: losing one point fails it.
W2 · What can leave the bench × dSPACE	4	2	3	4	4	**17**	Fits our HIL and CAN skills, but dSPACE already covers it. Our only edge is independent real-vehicle data. Exactly at threshold: any lost point fails it. Runs only if licence and bench access are confirmed by 15 Dec 2026.
W1 · Safety evidence for ML × Mobileye	4	3	3	2	4	**16**	Strong method fit, but R&D sits in Israel and Mobileye is not a sponsor. One warm channel would lift Access to 3 and the total to 17. First route: Stellantis engineers.
W2 · What can leave the bench × Applied Intuition	4	2	3	3	3	**15**	Applied builds on benches customers already own, so a cheaper bench adds nothing it lacks.
Zonal I/O & migration × dSPACE	3	2	3	2	3	**13**	dSPACE already tests zonal ECUs, and design or migration needs AUTOSAR and production work we lack.
Zonal I/O & migration × Applied Intuition	3	2	3	1	3	**12**	Migration happens inside multi-year Vehicle OS programs and requires ISO 26262.
OTA, commercial & specialty × dSPACE	3	2	2	2	2	**11**	dSPACE has no OTA product, and we lack UN R156 and fleet experience.
W2 · What can leave the bench × Mobileye	2	2	3	1	2	**10**	Its HIL is high-speed camera replay beyond our skills, built in-house, with no channel for us.
SBOM & provenance × dSPACE	2	1	2	2	3	**10**	No SBOM pain in its materials, and audit-grade evidence needs credibility we lack.
SBOM & provenance × Applied Intuition	2	1	3	2	2	**10**	Dedicated SBOM vendors exist, and our cyber background is coursework only.
OTA, commercial & specialty × Applied Intuition	2	1	3	1	2	**9**	Applied ships OTA inside Vehicle OS, so we would compete rather than partner. Open-source updaters also exist.

Pair	Fit	Diff	Need	Access	Strat	Total	What decides it
SBOM & provenance × Mobileye	2	1	3	1	2	**9**	Mobileye consumes compliance tools rather than selling them, and its cyber teams are out of reach.
Zonal I/O & migration × Mobileye	2	1	2	1	1	**7**	Mobileye is absent from zonal work: “zonal” does not appear in its 10-K.
OTA, commercial & specialty × Mobileye	2	1	2	1	1	**7**	Its OTA is component-level, and OEMs run their own on top.

CHAPTER 3 · WHERE WE FIT · HOW OUR THINKING CHANGED

New evidence changed our answer

We kept the first answer on the record. It was reasonable on what we knew on 5 Oct, and here is what moved us.

5 Oct Round 1 screen **13 Oct** Market claims audited **26 Oct** Company screen **30 Oct** BMCs v2

15 Nov First interview request **15 Dec** Access deadlines **End of Studio 1** Re-score with interviews

Round 1 · 5 Oct	After Round 2 · 26 Oct	What we learned
#1 HIL & virtual-ECU validation	**W2, conditional**: What can leave the bench	dSPACE owns HIL and virtual ECUs, and Applied’s HIL Sim runs on benches customers already own. Being good at HIL gives us empathy, not an advantage. Our only distinct asset is independent real-vehicle data. Starts only if licence and bench access are confirmed by 15 Dec 2026.
#2 Zonal migration of legacy ECUs	**Dropped**: Zonal migration	Migration lives inside OEM programs and needs AUTOSAR, ISO 26262 and freedom-from-interference evidence. dSPACE already tests zonal ECUs and Mobileye is absent. Best score: 13 of 25. Reopens with AUTOSAR and ISO 26262 training plus a Tier 1 sponsor. A test study might reach about 15, still below the bar.
#3 OTA for commercial & off-highway	**Dropped**: OTA for commercial & off-highway	Applied sells OTA inside Vehicle OS deals, Mender, RAUC and SWUpdate are open source, and we lack PKI, secure-boot and R156 experience. Best score: 11 of 25. Reopens with a specialty-OEM design partner and a certified update partner, or folded into the HIL work as OTA validation.
#4 Validation evidence, mid-size OEMs	**W1, first to test**: Safety evidence for ML components	All three companies publicly rely on simulation, scenarios and statistics for ML components. We also found no stated demand from mid-size OEMs, so the segment did not hold. The specific gap sits at large players. We redefined this position during the screen, so its pass is not like-for-like with Round 1.
#5 SBOM & cyber lifecycle evidence	**Still out**: SBOM & cyber evidence	The company screen agreed. None of the three sells SBOM tooling, so none pulls, and audit-grade evidence needs credibility we lack. Best score: 10 of 25. Reopens only with a cybersecurity faculty partner and a supplier design partner.
#6 Mixed-make fleet data	**Not reconsidered**: Mixed-make fleet data	The dossiers show no link at any of the three companies, and our research edge does not apply. It was not scored in Round 2. CAN and J1939 skills alone would not stand out in a crowded telematics market (team hypothesis).
#7 Edge-side data selection	**Only as a lead extension**: Edge-side data selection	Applied and dSPACE both describe long-tail data search as a pain. It could connect to our W1 work (our inference). It stands or falls with W1.

Why the reasoning changed

1. **We asked a sharper question.** Round 1 asked which positions suit our team and the market. Round 2 asked which real company we could reach, and what we could add that it lacks.
2. **We found evidence we didn't have.** Studying three companies showed that incumbents already hold most of the work behind HIL, zonal and OTA, and that all three rely on sampled testing for ML components in public.
3. **We counted an asset we had missed.** Round 1's capability list never named the verification research we can access through our faculty advisor. It became central to the strongest position.
4. **We raised the bar.** Round 2 added must-haves: we must be able to do the work and reach the company, and a score of mostly 3s no longer passes.

What stayed the same

- The market map and the thesis: the opening is pre-launch engineering work
- Integration and V&V remains the fastest-growing slice of auto software spend
- SBOM stays out, and our HIL skills stay useful

What we still don't know

- Whether anyone wants any of the three: the pains are inferred from public sources, and no one has confirmed them yet
- Whether our approach works on a realistic component within a build's time budget
- Whether dSPACE will act as a channel: sponsorship is not product interest

Round 1 priority (feasibility × market) and Round 2 totals use different criteria, so the numbers should not be compared directly. Both are team scores, not yet tested in interviews.

CHAPTER 3 · WHERE WE FIT · ENGINEERING PERSONAS

Six people we must convince

Ranked by who decides and who we can reach first.

Morgan Hale

dS-1 · Product Manager, AI code-generation · dSPACE

Decision-maker, interview #2. Decides whether new ML evidence enters the code-generation workflow.

"Simulation tells me the code matches the model. It doesn't tell me the model is safe to rely on." (imagined quote)

Full persona card

Jordan Pike

dS-2 · HIL Applications Engineer · dSPACE Inc., Wixom

Gatekeeper (access), interview #1. Our first contact, reachable through EcoCAR sponsorship. Can route us to product management.

"Half my week is explaining what has to stay on the bench and what can go virtual." (imagined quote)

Full persona card

Taylor Brandt

dS-3 · HIL / Virtual-Validation Lead · OEM using dSPACE

User / influencer, interview #3. The engineer who would actually use our evidence. Reachable via Stellantis, an EcoCAR sponsor.

"If it plugs into the chain I already run, I'll try it. If it's another island, it won't survive the next budget round." (imagined quote)

Full persona card

Rhea Calloway

AI-1 · Senior System Safety Engineer · Applied Intuition

Gatekeeper (veto), interview #4. Accepts or blocks new evidence types in the safety case.

"What I don't have is a sentence I can defend to an assessor about the inputs we never sampled." (imagined quote)

Full persona card

Daniel Ferreira

AI-2 · V&V Lead, Autonomy · Applied Intuition

Decision-maker (methods), interview #5. Decides which validation methods get adopted. Cares most about how long a check takes per build.

"Give me a number I can trace to a claim, and tell me how long it takes per build." (imagined quote)

Full persona card

Avery Lennox

ME-1 · Safety Assurance Expert · Mobileye

Influencer (method owner), interview #9. Near-miss persona (16 of 25, fails on access). Kept for when a warm introduction opens.

"We proved the policy. The open question is how much of the perception argument can be more than statistics." (imagined quote)

Full persona card

Every persona is a fictional archetype, not a real person, and every quote is illustrative. These replace the P1–P8 persona codes from Round 1. Start with field engineers: Applications and field engineers are the most reachable US contacts at all three companies. Win the safety gatekeeper: They hold a veto, so our output must speak in claims, assumptions and evidence. Sell evidence, not a tool: What gatekeepers need is evidence that fits their safety case. [49]

CHAPTER 4 · OUR WEDGES · WEDGE HYPOTHESES

Three wedges, each with a test that could kill it

Who has the problem, what we would offer, how big it could be, and how Studio 2 tests it. We describe our approach in general terms and keep the method private.

Focus area: integration, verification and validation (V&V) for software-defined vehicles, and specifically whether a model, a test or an update can be trusted. It is the fastest-growing slice of auto software spend, it gates every release, and it uses our HIL, CAV and research strengths. W1 and W2 came out of the company screen. W3 was added after it, in

the same focus area, and has not been scored against companies yet. No interview has tested any of the three, so all are low confidence.

W1. Safety evidence for ML components

First to test · Confidence: Low

Problem. A safety or validation engineer who must sign off a small ML component in an ADAS release can only show that the tested inputs passed, so the sign-off can be challenged. Better: evidence a reviewer accepts alongside scenario tests.

We believe safety and validation engineers signing off small ML components have had that evidence challenged in ways sampled testing cannot answer, and would accept stronger evidence delivered inside the toolchain they already run.

Where it came from: the company screen re-scoped ADAS validation evidence into this position. It scored 20 of 25 with dSPACE (pass), 19 with Applied Intuition and 16 with Mobileye.

End user	Safety and validation engineers signing off small ML components in ADAS and automated driving software
Pain	Simulation and statistics cover only the inputs sampled, so the evidence can still be challenged
Our narrow first foothold	One small learned component in a CAV perception stack
Input	A small ML component and a description of where it must work
Output	Evidence a safety reviewer can use alongside scenario tests
What changes for the user	Reviewers get something sampled testing alone cannot provide
Workaround today	Scenario sweeps, perturbation tests and statistics; MIL/SIL/PIL simulation
Budget owner	Still to find. First guesses: a tool product manager (dS-1) or a V&V methods lead (AI-2)
Part of EcoCAR it addresses	Driverless Mobility (GM). CAV work, K CAV Perception Simulation Demonstration. Applies if our perception stack includes a small learned component. Uses simulation data only.

Persona pains this wedge answers

Persona	Their pain	How this wedge answers it
AI-1	No sentence to defend to an assessor about the inputs that were never sampled	Evidence that speaks to more than the sampled inputs, stated in safety-case terms
dS-1	Simulation shows the code matches the model, not that the model is safe; customers must show auditable AI under ISO/PAS 8800	Evidence that fits the tool flow customers already run
AI-2	Wants a number that traces to a claim, and to know how long it takes per build	Each result tied to a stated claim, with run time logged
dS-3	Will not adopt another tool island outside the chain already in use	Delivered inside the existing toolchain

Weakest technical assumption: That useful evidence can be produced for a realistic component within a per-build time budget. If it only works on very small components, W1 fails. We test this with a prototype and a run-time log by the end of February 2027.

Early top-down market proxy

Level	Arithmetic	2025	2035
TAM proxy	ADAS/AD software [1]	about $9B	about $50B
SAM proxy	TAM x 5–10% spent on validation and safety-evidence tooling x 10–20% for ML components in scope	$45M–$180M	$250M–$1B

All percentages are our assumptions. No named automotive customer turned up for this kind of evidence, so the SAM could be near zero today.

What is still missing

- No bottom-up estimate yet: we do not know how many programs would pay, or how much.
- The 5–10% and 10–20% shares in the SAM are our assumptions, not sourced figures.
- No buyer or budget line has been named by anyone outside the team.

Technical assumptions

- Our approach applies to the type of component we choose. Scope is being confirmed.
- Evidence can be produced within a per-build time budget.
- It will not cover a full camera-to-control stack.
- Standards permit this kind of evidence but do not require it.
- Early output is research evidence, not qualified under ISO 26262-8.

Studio 2 validation plan

Hypothesis	Type	Test	Fails if	Evidence source
Gatekeepers see a gap that statistics do not fill	Desirability	6 safety or V&V interviews: "When was evidence for an ML component last challenged, and by whom?"	Fewer than 3 of 6 describe a concrete recent case	Interview notes, coded supported, refuted or unclear in the Registry
We can do this on a realistic component	Feasibility	Build a prototype on one small component and log run time	Not done by end of Feb 2027 within the stated per-build time, or 1 hour if none is stated	Prototype run-time log on a public simulation component
A tool vendor could act as a channel	Viability	Short demo to a product contact after an introduction	They decline a second meeting or name qualification as a hard blocker	Meeting notes and follow-up emails
Someone would pay or commit	Viability	Ask who funded the last new kind of safety evidence, then ask for a next step	Nobody names a budget or offers time, reputation or money	Interview notes: a named budget owner or a written next step

Personas: dS-1, dS-2, dS-3, AI-1, AI-2. **Weakest link:** Demand: stronger evidence is absent today, but nobody has been shown asking for it.

W2. What can leave the bench

Weakest today · Confidence: Low

Problem. A HIL or virtual-validation lead deciding whether a test can move from a scarce HIL bench to a virtual ECU has no measured data on what the virtual run misses. Better: a data-backed rule for which tests must stay on the bench.

We believe HIL and virtual-validation leads have recently kept, repeated or delayed a test because no measured data showed what a virtual ECU would miss, and would use

independent, measured deviations to decide which tests can move off the bench.

Where it came from: the company screen narrowed our Round 1 #1, HIL & virtual-ECU validation, to this question. It scored 17 of 25 with dSPACE, a conditional pass.

End user	HIL and virtual-validation leads
Pain	Tests move from HIL to virtual ECUs without measured data on what is lost
Our narrow first foothold	One controller function compared across virtual ECU, HIL and vehicle
Input	The same test cases run in each environment
Output	Measured CAN timing, I/O and fault-response differences, plus a “what must stay on HIL” checklist
What changes for the user	Decisions about where a test runs rest on data
Workaround today	Keeping tests on HIL by judgment (untested); vendor SIL–HIL co-simulation and virtual-ECU services
Budget owner	Still to find: a Studio 2 question
Part of EcoCAR it addresses	Driverless Mobility (GM). Closest Year 1 link is G-3.2 Vehicle Data Simulation, which compares the virtual model with Blazer data and explains discrepancies. GM Year 1 deliverables do not mention HIL, so bench work would be extra. Vehicle-derived data stays out of innovation-track material.

Persona pains this wedge answers

Persona	Their pain	How this wedge answers it
dS-2	Spends half the week explaining what has to stay on the bench and what can go virtual	Measured differences to point to
dS-3	Will not adopt another tool island outside the chain already in use	Results from the same test cases and tools the team already uses

> **Weakest technical assumption:** That the differences between virtual ECU, HIL and vehicle are large and repeatable enough to change where a test runs. If they are small or random, the study changes no decision. We test this by running one controller function three ways once licences are confirmed.

Early top-down market proxy

Level	Arithmetic	2025	2030
TAM proxy	HIL [55] $1.10B + virtual-ECU software [56] $0.95B (the two may overlap)	about $2.05B	not summed
SAM proxy	Virtual-ECU spend at fidelity levels 3 and 4 [56]	about $398M	about $1.09B
Wedge	SAM x 2–5% attach rate for fidelity assurance	$8M–$20M	$22M–$54M

The attach rate is our assumption and could be zero if vendors bundle this. Regulators add a second reason to care: UNECE work calls a harmonized simulation-credibility framework “essential” [59].

What is still missing

- The 2–5% attach rate is our assumption.
- The virtual-ECU figures come from one low-transparency source; its “levels” describe how closely a virtual ECU matches the real one, not driving automation.

- The HIL and virtual-ECU market figures may overlap, so we do not add them for 2030.

Technical assumptions

- One controller function can run in all three environments with comparable test cases.
- The differences are large enough to change where a test runs.
- One controller on one vehicle generalizes enough to interest anyone else.
- Sponsorship terms cover the licences and bench time this needs.

Studio 2 validation plan

Hypothesis	Type	Test	Fails if	Evidence source
Leads lack data on what must stay on HIL	Desirability	4 HIL interviews: "Walk me through the last time you moved a test from HIL to SIL. What made you trust it?"	Fewer than 2 of 4 report a recent decision where fidelity data was missing	Interview notes, coded in the Registry
A one-controller study is useful	Desirability	Show a one-page mock result after the pain questions are finished	They say it would not change any customer conversation or decision	Reactions to the mock result, in interview notes
We can get tools and bench time	Feasibility	Confirm licence scope and bench access through the sponsor liaison	Not confirmed by 15 Dec 2026	Written confirmation from the sponsor liaison
Someone would pay or commit	Viability	Ask who funded the last comparable study, then ask for a next step	Nobody names a budget or offers time, reputation or money	Interview notes: a named budget owner or a written next step

Personas: dS-2, dS-3. **Weakest link:** Payment: this may be a publication, not a product, and an OEM could run the study itself.

W3. Model-update re-test scoping Newest · Confidence: Low

Problem. An integration or validation lead releasing a retrained perception model must decide what to re-test, and either re-runs nearly everything or cuts scope on judgment. Better: a smaller re-test scope that comes with recorded reasons a reviewer accepts.

We believe integration and validation leads who retrain perception models either re-run far more testing than a change needs or cut scope on judgment they struggle to defend, and would adopt guidance on what to re-test, with the reasoning recorded.

Where it came from: added after the company screen, inside the same V&V focus area, by asking what else must be trusted after a change. It was not scored against companies. It ties most directly to our GM CAV deliverables (K and D).

End user	Integration and validation leads shipping ADAS updates
Pain	Software-defined vehicles keep changing after launch. After a model is retrained, code-based change tools cannot show which behaviors moved
Our narrow first foothold	Retrained perception models
Input	A new model version and the team's existing test set
Output	Guidance on what to re-test, with the reasoning recorded

What changes for the user	Less wasted re-testing and a recorded reason for the scope chosen
Workaround today	Re-running everything, or choosing by judgment (inferred; to test)
Budget owner	Still to find: a validation director or software release lead
Part of EcoCAR it addresses	Driverless Mobility (GM). CAV work, K CAV Perception Simulation Demonstration (teams must iteratively expand, refine and validate perception) and D CAV Toolchain Check (version control process).

Persona pains this wedge answers

Persona	Their pain	How this wedge answers it
AI-2	Wants to know how long checks take per build	A smaller, reasoned re-test scope for each model update
dS-3	Budget pressure on anything outside the chain already in use	Fewer re-runs inside the existing toolchain
Gap	No persona yet for a perception-model release lead, the likely main user	We will draft one from public job postings (see Our gaps)

Weakest technical assumption: That a retrained model's change can be analysed well enough to beat simple baselines: re-running everything, random selection and a published method. If it cannot, W3 adds nothing. We test this with a simulation benchmark with known faults.

Early top-down market proxy

Level	Arithmetic	2025	2035
TAM proxy	ADAS/AD software [1] (shared with W1, so never add the two)	about $9B	about $50B
SAM proxy	TAM x 5–10% spent on validation tooling x 20–40% spent re-testing after changes	$90M–$360M	$500M–$2B

All percentages are our assumptions. A separate bottom-up view: 30–60 ADAS programs at $100k–$300k a year gives $3M–$18M a year. In QNX's 2026 vendor-commissioned survey of 1,100 developers, 44% named debugging and testing their top challenge (36% in its 2025 survey) [57].

What is still missing

- The top-down SAM ($90M–$360M) and the bottom-up view ($3M–$18M a year) differ by 5 to 100 times. We trust the bottom-up more for a first product, because the share of validation spend that goes to re-testing is our assumption.
- No price point has been tested.
- No buyer or budget line has been named by anyone outside the team.

Technical assumptions

- A model change can be analysed well enough to guide re-testing.
- It must beat simple baselines. Published selection methods have cut ADAS road-test suites by 89% while keeping 79% of failures [58].
- Simulation is a fair first stand-in for real release data.
- Ranking is useful even if safety rules prevent skipping any test.

Studio 2 validation plan

Hypothesis	Type	Test	Fails if	Evidence source
Scoping re-tests is a costly problem	Desirability	6 interviews: “When one function last changed, how did you decide what to re-test?”	Fewer than 3 of 6 describe a costly or disputed case from the past 12 months	Interview notes, coded in the Registry
Current tools do not settle it	Desirability	“What have you tried, bought or built in the past year to deal with that?”	4 of 6 say existing tools or in-house methods already solve it	Interview notes; named tools checked against public product pages
Reviewers accept a documented scope	Desirability	“When was a test last left out of a re-run, and who agreed?”	No sign-off role has accepted a reduced scope with written reasoning	Interview notes from safety and sign-off roles
It finds problems with less testing	Feasibility	Simulation benchmark with known faults, compared with running everything, random selection and a published method	By end of Feb 2027, cannot catch at least 90% within half of the tests	Benchmark results on a public simulation dataset
Someone would pay or commit	Viability	Ask who funds re-test work today, then ask for a next step	Nobody names a budget or offers time, reputation or money	Interview notes: a named budget owner or a written next step

Personas: AI-2, dS-3 and the validation engineers on our stakeholder maps. **Weakest link:** Build vs. buy: teams may already solve this in-house, and safety reviewers may refuse any reduced test scope.

DFV scoring

Criterion (weight)	W1	W2	W3	Basis
DESIRABILITY				
Market growth 1	4	4	4	ADAS/AD software about 19% a year; virtual ECU about 18%, HIL about 10%
Market scale 0.75	3	3	3	Many validation teams, concentrated in large programs
Dissatisfaction with current state 0.75	3	3	3	Pain is documented but inconsistently stated; W2’s is mostly vendor-written
Importance to end user 0.8	4	3	3	W1 ties to a release gate; W2 and W3 mainly affect schedule and cost
Evidence of workaround 0.95	3	3	3	Recurring workarounds are visible; W3’s is inferred
FEASIBILITY				
Strategic assets 0.8	3	3	3	Research access, bench and simulation all apply, but each has open conditions
Key activities 0.8	3	2	2	W1’s process is not yet built; W2 is replicable; W3 has published methods to beat
Strategic network 0.7	3	4	3	dSPACE sponsorship is closest to W2; no contact has engaged yet
VIABILITY				
Strategic priority 1	4	3	4	Auditable AI and software-update management are stated priorities; dSPACE already ships SIL–HIL co-simulation
Ownership and governance 0.9	2	2	2	A likely user exists, but no budget owner is known
Long-term optionality 0.85	4	2	4	W1 and W3 could share reusable capability; W2 is closer to a single study

Criterion (weight)	W1	W2	W3	Basis
Low cost base 0.65	3	2	3	Software with per-customer setup; W2 needs bench and vehicle time
Willingness to pay 0.7	2	2	2	No source shows anyone paying for any of these
Payback 0.5	2	2	2	Depends on adoption we cannot yet show

Result	W1	W2	W3	Max
Desirability	14.55	13.75	13.75	21.25
Feasibility	6.9	6.8	6.1	11.5
Viability	13.55	10.2	13.55	23
Total	**35**	**30.75**	**33.4**	**55.75**
Position in range	64%	57%	61%	Middle: 45–70%

Pre-interview desk scores from 1 to 5, using the fall-workshop DFV tool's 14 criteria and suggested weights [54]. All three land in its Middle band, so the scores do not yet pick a winner. W1 and W3 are close enough that two one-point changes would erase the gap. Dissatisfaction, workaround, ownership, willingness to pay and payback need interview evidence. W2 feels weakest as a business today. W1 and W3 test one idea from two sides, with different buyers and jobs, so a weak approach would hurt both. W3 must therefore beat a simple baseline on its own.

CHAPTER 5 · NEXT STEPS · OUTREACH PLAN

Who we will interview first, and how we reach them

#	Persona	Who we look for	Wedges	Channel	Status and next step
1	**dS-2**	HIL applications engineer, dSPACE Inc. (Wixom, MI)	W2, W1	EcoCAR sponsor contact Sponsor	In progress First request by 15 Nov
2	**dS-1**	Product manager, AI code generation, dSPACE	W1	Warm introduction from dS-2 Sponsor	In progress After the first dSPACE call
3	**dS-3**	HIL or virtual-validation lead at an OEM using dSPACE, such as Stellantis	W2, W3, W1	Stellantis sponsor contact; WMU alumni Both	In progress Finish the LinkedIn and alumni search
4	**AI-1**	Senior system safety engineer, Applied Intuition	W1	WMU alumni; LinkedIn message; conference talks Independent	In progress Named contact needed by 15 Dec
5	**AI-2**	V&V lead, autonomy, Applied Intuition	W1, W3	WMU alumni; LinkedIn message Independent	In progress Named contact needed by 15 Dec
6	**New**	Perception or ADAS validation lead who retrains models, at an OEM or Tier 1	W3	WMU alumni; LinkedIn; SAE and IEEE events Independent	Not started Draft the persona, then search
7	**ME-1**	Safety assurance expert, Mobileye	W1	Warm introduction only, through Stellantis engineers Independent	Parked Waits for a warm channel

Aim: About 10 people, covering 6 W1, 4 W2 and 6 W3 conversations (many speak to two wedges), with licence and data access confirmed by 15 Dec. Names and contact details stay in our private tracker, not on this public site. Five of

the seven rows use an independent channel, so at least half of our interviews can come from independent outreach, as Studio 2 requires.

How we reach out: Ask for 20–30 minutes about their past work · Say who we are and why we are asking · Ask before taking notes; never record without permission · Never ask for confidential OEM or supplier information · At most two polite follow-ups · Offer our summary back.

How we interview: Ask about the past, not the future · Don't pitch · Dig into workarounds · Quantify · Record what disproves us.

After each interview: update the persona card, mark each assumption supported, refuted or unclear in the Hypothesis & Assumptions Registry, and re-score every pair. Surviving candidates update the B-2.3.4 wedge hypotheses.

CHAPTER 5 · NEXT STEPS · INNOVATION IN SOCIETY

What could go wrong for people, and what we do about it

	What could go wrong	What we do about it	Still open
Safety	Our evidence is read as proof that an ML component is safe, or W3 skips a test that mattered.	Every output states its scope and limits. W3 ranks tests and never drops one without sign-off. Nothing we build is qualified under ISO 26262, and we say so.	How safety assessors want limits stated
Security	Customer models and test sets are valuable IP, and a new tool could open a way into a build pipeline.	Year 1 uses only public or simulation models. Any later design keeps customer models in the customer's environment.	What security review a vendor needs before a pilot
Privacy	Driving data can identify people, as the FTC order against GM shows.	W1 and W3 use simulation data only. W2 uses controller signals, not driver data.	Interview notes hold names: private tracker, shared only with consent
Fairness	Perception can work worse for children, wheelchair users, at night or in rain. Averages hide this.	Include these people and conditions in W1's checks and W3's benchmarks, and report results by condition.	No public dataset with this coverage chosen yet
Environment	Simulation and re-testing use computing energy; benches and test drives use hardware.	W2 and W3 aim to cut bench time and re-runs that are not needed. We log computing time per run.	No energy estimate yet

Every market claim is sourced and labelled in the evidence audit. Public sources only: Rev 5 does not permit OEM data in this deliverable. Personas are fictional, and their quotes are imagined.

CHAPTER 5 · NEXT STEPS · RISKS & GAPS

What could prove us wrong, and what we do about it

Demand

0 of 3

wedge pains confirmed by anyone outside the team

What we do: First interview request to the dSPACE applications engineer; 2–3 exploratory interviews before Studio 1 closes **By 15 Nov.**

Fit with our vehicle

W1

counts as vehicle work only if our CAV perception stack has a small learned component

What we do: Get a yes or no from our CAV lead **By Check-In.**

Access

15 Dec

deadline for dSPACE licence scope and a named Applied Intuition contact, or W2 and one W1 pair fail

What we do: Finish the persona-matched search and log channel and status for each person **By 15 Dec.**

Build

Feb 2027

W1 and W3 prototypes must work on a realistic component in simulation, or we drop them

What we do: Prototype both on public simulation components and log run time **By Feb 2027.**

Smaller gaps we are closing

Area	Gap	Action	By
Personas	No persona yet for W3's likely user, a perception-model release lead	Draft one from public job postings, like the other six	30 Nov
Evidence	Some W1 evidence cites a team report we keep private	Cite the public job postings and company pages directly	30 Nov
Market	W1 and W2 have only top-down proxies, and W3's two estimates disagree	Build bottom-up counts of programs and prices; ask about price in interviews	Studio 2
Sources	Nine market facts still need checking against the original source	Check each one by hand and update the audit	30 Nov
Process	No Hypothesis & Assumptions Registry yet	Move each wedge's assumptions and tests into the registry template	30 Nov

What we still don't know

- Whether anyone wants any of the three: the pains are inferred from public sources, and no one has confirmed them yet
- Whether our approach works on a realistic component within a build's time budget

- Whether dSPACE will act as a channel: sponsorship is not product interest

CHAPTER 5 · NEXT STEPS · QUESTIONS FOR YOU

Where your advice would change what we do next

1. Does W1 count as “closely related to vehicle development” if it applies to one learned perception component?
2. Could you introduce us to dSPACE product management, and to GM ADAS or safety engineers?
3. Which wedge would you test first, and what evidence would convince you to drop one?
4. Is a top-down market proxy enough at this stage, or should every wedge have a bottom-up estimate?
5. How do sponsor IP agreements apply when a wedge builds on faculty research?

APPENDIX · ROUND 1 SCORING

From seven open pockets to three targets

Priority = team feasibility (skill, assets, prototype, access, edge) × market attractiveness from the B-2.3.1 map.

8 value-chain layers mapped
B-2.3.1 · Six stacked layers plus two cross-cutting (cybersecurity; integration & V&V)

7 open pockets found
B-2.3.1 · One critical, not-yet-crowded pocket per layer; none in software platforms

4 shortlisted
B-2.3.1 · Pockets in the critical-and-fragmented quadrant of the 11-area matrix

3 Round 1 targets
B-2.3.3 · All 7 pockets re-scored for team feasibility; 3 of the 4 shortlisted won, SBOM fell to #5

2 after the company screen
B-2.3.3 · Round 2 tested these against real companies. They became W1 and W2; W3 was added after the screen (B-2.3.4)

Position	Skill fit 25%	Asset leverage 20%	Prototype Yr 1–2 20%	Stakeholder access 15%	Defensible edge 20%	Feasibility	Market	Priority
1. HIL & virtual-ECU validation ★	5	5	5	5	4	**4.80**	4.4	**21.1**
2. Zonal migration of legacy ECUs ★	5	5	4	4	4	**4.45**	4.5	**20.0**
3. OTA for commercial & off-highway ★	4	3	3	4	3	**3.40**	4.2	**14.3**

Position	Skill fit 25%	Asset leverage 20%	Prototype Yr 1–2 20%	Stakeholder access 15%	Defensible edge 20%	Feasibility	Market	Priority
4. Validation evidence, mid-size OEMs	4	4	3	3	3	**3.45**	3.5	**12.1**
5. SBOM & cyber lifecycle evidence ★	3	2	4	3	2	**2.80**	4.0	**11.2**
6. Mixed-make fleet data	4	3	4	4	2	**3.40**	3.0	**10.2**
7. Edge-side data selection	4	4	4	3	3	**3.65**	2.6	**9.5**

★ shortlisted in B-2.3.1. All seven open pockets were re-scored, so the shortlist could be overturned. Draft scores for the team to calibrate in a 30-minute scoring session, and to re-check after Studio 1 interviews [46]. Skills: controls & HIL testing (model-based design); embedded & ECU software (C/C++, CAN, AUTOSAR); CAV/ADAS & perception; cybersecurity, cloud & data. Assets: HIL bench & test lab; test vehicle with E/E network access; sponsor & mentor contacts; fleet & commercial-vehicle contacts.

How market attractiveness was judged

Market attractiveness (1–5) is the team's judgement from the B-2.3.1 map: criticality to launch or approval, combined with how open the competition is.

Position	Criticality to launch or approval	Competition	Market
HIL & virtual-ECU validation	**High**: Validation evidence gates every software release (ISO 26262)	**Low**: Tool market is fragmented below the leaders (Applied Intuition, dSPACE, Vector)	4.4
Zonal migration of legacy ECUs	**High**: Every zonal platform must port and re-validate legacy functions before SOP	**Low**: An open pocket in a crowded layer: SoC vendors fight over central compute, few serve migration	4.5
OTA for commercial & off-highway	**High**: UN R156 software-update management is required for EU type approval	**Low**: Large OEMs insource; Sibros, Excelfore and Harman serve only parts of this market	4.2
Validation evidence, mid-size OEMs	**High**: Safety evidence is needed for every ADAS release	**High**: Sits in the crowded ADAS layer, next to well-funded L2+ stacks	3.5
SBOM & cyber lifecycle evidence	**High**: UN R155 type approval and the US Commerce rule make provenance a gate	**Moderate**: Specialists (Upstream, VicOne, PlaxidityX) and SBOM vendors already compete	4.0
Mixed-make fleet data	**Medium**: Fleets need it for uptime and cost, but no launch or approval depends on it	**High**: Crowded by telematics incumbents (Samsara, Motive, Geotab)	3.0
Edge-side data selection	**Low**: Cuts data cost, but no launch or approval depends on it	**Low**: Few vendors, because the market is still unproven	2.6

How sensitive is the ranking?

Weights: across 10 single-weight changes of ±10 points, the top three hold in 10 and HIL stays #1 in 10.

Scores: across 78 single-score changes (±1 on a criterion, ±0.5 on market), the top three hold in 78. #1 is the close call (21.1 vs 20.0): HIL skill fit −1, HIL market −0.5 or Zonal market +0.5 would put Zonal first.

APPENDIX · DOSSIERS

Where W1 and W2 came from

Confidence rubric: High needs external interviews and primary data, so no target can be High before Studio 1.

W1, first to test · Safety evidence for ML components

dSPACE 20/25 (Pass) · Applied Intuition 19/25 (If we find a contact) · Mobileye 16/25 (Near-miss) · evidence confidence: **Low**. The key assumption, that safety engineers want this, is untested. Public sources only so far.

All three companies validate ML components with simulation, scenarios and statistics, which show that the tested inputs pass. They say nothing about the inputs never sampled. Applied's safety roles ask for ISO/PAS 8800, UL 4600 and "safety of ML systems", and dSPACE takes part in ISO/PAS 8800 [49].

Our edge: Our team has access to verification research through our faculty advisor. We are keeping the method private for now. Our edge would be turning that research into evidence a safety team can use, shown with sponsor tools [49].

Users

Safety and validation engineers signing off small ML components in ADAS and automated driving software (closest personas: AI-1, dS-3)

Budget owners

Still to find. First guesses: a tool product manager (dS-1) or a V&V methods lead (AI-2)

Gatekeepers

AI-1 Applied system safety engineer (veto over evidence types); dS-2 dSPACE applications engineer (access)

Influencers

ISO/PAS 8800 contributors; ME-1 Mobileye safety assurance expert (near-miss)

Workaround today

Scenario sweeps, perturbation tests and statistics; MIL/SIL/PIL simulation

EVIDENCE FOR

- All three companies publicly rely on simulation, scenarios and statistics for ML components [49]
- dSPACE's public materials describe MIL/SIL/PIL simulation checks for its neural-network workflow [49]
- Applied's safety postings ask for ISO/PAS 8800, UL 4600 and ML-safety skills [49]
- dSPACE sponsors the 2026 challenge and has a US office in Wixom, MI [49]

EVIDENCE AGAINST

- Absent is not the same as wanted: no source shows anyone asking for this. Need scores only 3 of 5 [49]
- Related analysis tools are becoming widely available, which lowers everyone's build cost [49]
- It will not cover a full camera-to-control stack, and our approach is not qualified under ISO 26262-8 [49]
- Absence in public sources is not proof that nothing exists internally [49]

FIRST TEST

Interview the dSPACE applications engineer, then the product manager: What do customers ask you about verifying ML components on the bench, if anything? When customers ask how a generated network is verified, what do you show them today? What other verification methods have customers or your team tried?

WE DROP IT IF

- Fewer than 3 of 6 safety or V&V interviewees describe a recent case where ML evidence was challenged
- By end of Feb 2027, the prototype cannot handle a realistic network within the per-build time budget
- dSPACE product management declines a second meeting after the demo

OPEN COMMERCIAL QUESTIONS

- A feature inside the dSPACE toolchain, or a stand-alone evidence service?
- Who pays: the tool vendor's product team or the OEM's safety team?

- Early TAM and SAM proxies are on the Wedge hypotheses slide.

W2, conditional · What can leave the bench

dSPACE 17/25 (Conditional, deferred) · Applied Intuition 15/25 (Fail) · Mobileye 10/25 (Fail) · evidence confidence: **Low**. Access is unconfirmed, and the pain evidence comes mainly from the vendor.

Moving tests from HIL to virtual ECUs saves bench time, but dSPACE's own SIL lead writes that timing and interrupt effects often appear only at HIL or vehicle level. We think small teams and mid-size customers lack public data for deciding what must stay on the bench [49].

Our edge: Our only distinct asset is independent real-vehicle data. We would run one EcoCAR controller function on a virtual ECU, on HIL and in the vehicle, measure CAN timing, I/O and fault-response differences, and publish a "what must stay on HIL" checklist. This is a weak advantage, so we pursue it only alongside W1 [49].

Users
HIL and virtual-validation leads (dS-3); zonal HIL engineers

Budget owners
Still to find: a Studio 2 question

Gatekeepers
dS-2 dSPACE applications engineer (access); EcoCAR sponsor liaison (licences)

Influencers
dSPACE Team Lead SIL E-Mobility; dSPACE V-ECU Task Force

Workaround today
Keeping tests on HIL by judgment (untested); vendor SIL–HIL co-simulation and virtual-ECU services

EVIDENCE FOR

- Our skills match dSPACE's US HIL applications-engineer role, and EcoCAR teams have used dSPACE HIL [49]
- It is core to dSPACE: SIL–HIL co-simulation launched in Jun 2026 [49]
- The same bench could support our W1 work [49]

EVIDENCE AGAINST

- dSPACE already sells entry HIL and virtual-ECU services, so Differentiation scores 2 of 5 [49]
- The pain evidence is mostly dSPACE-authored, and no source shows demand for independent fidelity data [49]
- At 17 of 25 it sits exactly on the threshold: any lost point fails it [49]

FIRST TEST

Interview the dSPACE applications engineer and an OEM validation lead: When did a customer last find a problem on HIL that SIL had missed? Walk me through the last time you moved a test from HIL to SIL. What made you trust it?

WE DROP IT IF

- Fewer than 2 of 4 HIL interviewees report a recent decision where fidelity data was missing
- The dSPACE engineer or OEM lead says the result would not change any customer decision
- No licence or bench access confirmed by 15 Dec 2026

OPEN COMMERCIAL QUESTIONS

- A public checklist that builds credibility, or a paid study?
- Does it stand alone, or only support W1 with dSPACE?
- Early TAM and SAM proxies are on the Wedge hypotheses slide.

Earlier targets, kept on record

Round 1 dossiers as written on 5 Oct, each with what changed.

Round 1 #1 · HIL & virtual-ECU validation

Feasibility 4.80/5 × market 4.4 = priority 21.1 · evidence confidence: **Medium**. Survey and market evidence plus first-hand experience; no external interviews yet.

> **What changed in Round 2: narrowed.** dSPACE already owns HIL and virtual ECUs, and Applied's HIL Sim runs on benches customers already own. Best score 17 of 25, exactly at the threshold. We narrowed this to W2, What can leave the bench.

The most trusted tests run on HIL benches, which are expensive, scarce and slow to set up, so teams queue for bench time. Virtual ECUs could run tests faster, but engineers and safety assessors often don't trust their timing and I/O fidelity.

Our edge: We run HIL benches ourselves, can prototype tooling without safety certification, and have sponsors in the test-tool world. The niche: make scarce bench time go further and judge when a virtual-ECU result can be trusted, not replace dSPACE or Applied Intuition.

Users

Test-Bench Engineer; Embedded CI Engineer

Budget owners

HIL Test Lead; Director of Validation

Gatekeepers

Functional-safety assessor (will virtual results count as evidence?); purchasing and tool approval

Influencers

dSPACE, Vector, Applied Intuition

Workaround today

Spreadsheets for bench scheduling; homemade test scripts; HIL suites from dSPACE or Vector

EVIDENCE FOR

- Integration and V&V services are the fastest-growing slice of auto software spend: 11.4%/yr to 2035 [1]
- Developers cite long cycles (37%) and debugging and testing (36%) as top problems (QNX 2025 vendor survey, n = 1,100) [35]
- Our own HIL engineers live this bottleneck: useful empathy, not proof of the market [47]

EVIDENCE AGAINST

- Well-funded incumbents: Applied Intuition was valued at $15B in 2025 and says it serves 18 of the top 20 OEMs [32]; Synopsys now includes Ansys simulation [43]
- The 52% engineering-VP figure is a vendor byline with no published sample size [36]
- Many OEMs run in-house HIL labs, and a safety assessor can veto virtual evidence (untested) [47]

FIRST TEST

Interview a HIL test lead and a test-bench engineer: How long do teams wait for a bench? Out of the last ten failed tests, how many were real software bugs?

WE DROP IT IF

- Benches aren't actually a bottleneck: teams have plenty of capacity
- Virtual ECUs are already trusted and widely used in their organisation

OPEN COMMERCIAL QUESTIONS

- Would they buy software, a managed validation service, or more bench capacity?

- Who owns the budget: the validation director or the software platform team?
- What does one week of bench queue cost a vehicle program?

Round 1 #2 · Zonal migration of legacy ECUs

Feasibility 4.45/5 × market 4.5 = priority 20.0 · evidence confidence: **Medium**. Strong long-run direction, but adoption timing and port-versus-rewrite are open questions.

> **What changed in Round 2: dropped.** Migration lives inside OEM programs and needs AUTOSAR, ISO 26262 and freedom-from-interference evidence. dSPACE already tests zonal ECUs and Mobileye is absent. Best score 13 of 25. Reopens with AUTOSAR and ISO 26262 training plus a Tier 1 sponsor.

Moving to zonal architecture means splitting, porting and re-validating dozens of legacy functions onto new zonal controllers and central computers: remapping signals, re-checking timing, separating safety-critical from non-critical functions, and retesting.

Our edge: Our test vehicle's E/E network plus our HIL bench lets us migrate one real legacy function onto a zonal-style controller and re-validate it end to end, a demo few startups can show. A full migration is bigger than a student project.

Users

Systems Integration Engineer; Tier 1 Zonal Software Lead

Budget owners

E/E Architect; vehicle chief engineer

Gatekeepers

Functional safety (ISO 26262); program timing (the SOP date)

Influencers

Tier 1s (Bosch, Aptiv); SoC vendors (NVIDIA, Qualcomm)

Workaround today

E/E architecture tools, DBC/ARXML signal databases, Simulink and spreadsheets for effort estimates

EVIDENCE FOR

- McKinsey expects 77% of E/E architectures to be zonal or central by 2035, up from under 30% today [1]
- Zone control units grow from $4B to $13B over 2030–35 [1]
- Consolidation is real: Rivian's Gen 2 R1 cut ECUs from 17 to 7 [11]

EVIDENCE AGAINST

- Timing is uncertain: in an NXP survey only about a third of OEM respondents expect body-zonal to be the main production architecture by 2030 [45]
- Programs can stall: Ford scrapped its FNV4 zonal architecture in 2025 [38]
- Tier 1s already do much of the migration work, and OEMs may rewrite rather than port (untested) [47]

FIRST TEST

Interview an E/E architect and a Tier 1 zonal lead: How do you estimate the effort to move one legacy function? On your last program, how much was ported versus written new?

WE DROP IT IF

- Migration is mostly greenfield: new software, not ported
- Effort estimation is already reliable, or porting is largely automated or cheaply outsourced

OPEN COMMERCIAL QUESTIONS

- Is the buyer the OEM architecture team or the Tier 1 doing the port?
- A dependency-analysis tool, or an engineering service per migration?

- Is demand a burst per vehicle program, or recurring across model years?

Round 1 #3 · OTA for commercial & off-highway

Feasibility 3.40/5 × market 4.2 = priority 14.3 · evidence confidence: **Low**. Regulatory pull is clear, but vendors exist and our assets are weakest here.

> **What changed in Round 2: dropped.** Applied sells OTA inside Vehicle OS deals, open-source updaters exist, and we lack PKI, secure-boot and R156 experience. Best score 11 of 25. Round 1's Low confidence proved right. Reopens with a specialty-OEM design partner and a certified update partner, or folded into the HIL work as OTA validation.

Large passenger-car OEMs build their own OTA. Commercial, specialty and off-highway makers usually lack the scale: their vehicles carry ECUs from many suppliers, often without A/B memory, on J1939 networks, so an update often means a technician or dealer visit.

Our edge: Embedded and cloud skills plus direct fleet and commercial-vehicle contacts for interviews and pilots. Limitation: no commercial test vehicle yet, which is why this ranks #3.

Users

CV Embedded Lead; Fleet Maintenance Manager

Budget owners

Engineering director at the CV maker; fleet operations director

Gatekeepers

Homologation (UN R156); cybersecurity (UN R155)

Influencers

Telematics providers (Motive, Samsara), Harman; dealers

Workaround today

Supplier-specific flashing tools, dealer service tools, spreadsheets that track software versions

EVIDENCE FOR

- UN R156 requires a managed software-update process for EU type approval [26]
- Only ~15% of recalled US vehicles since 2022 were eligible for an OTA fix, so most still need a physical visit [3]
- 891 US recalls covered 29.3M vehicles in 2025 [2]

EVIDENCE AGAINST

- Specialists already serve parts of this market: Excelfore reports 20+ automaker customers [41], alongside Sibros and Harman
- Fleet telematics incumbents are already on the vehicle and could add OTA: Samsara runs $2.1B ARR [23]
- No commercial test vehicle, and ECU suppliers may refuse bootloader access (untested) [47]

FIRST TEST

Interview a CV embedded lead and a fleet maintenance manager: How does a software fix reach a vehicle already in the field today? How many vehicle-days were lost to updates and recalls last year?

WE DROP IT IF

- They already use a vendor OTA solution that works
- Software-related downtime is negligible compared with mechanical downtime

OPEN COMMERCIAL QUESTIONS

- Per-vehicle licence, per-campaign fee, or bundled with telematics?

- Does fleet demand pull strongly enough to make the CV maker's business case?
- Partner with telematics providers as a channel, or compete with them?

Screened out in Round 1 · SBOM & cyber lifecycle evidence

Feasibility 2.80/5 × market 4.0 = priority 11.2 · evidence confidence: **Screened out**. Real need, weak fit: team feasibility 2.80 of 5.

> **What changed in Round 2: still out.** The company screen reached the same answer. None of the three sells SBOM tooling, so none pulls, and audit-grade evidence needs credibility we lack. Best score 10 of 25. Reopens only with a cybersecurity faculty partner and a supplier design partner.

UN R155 type approval and the US Commerce connected-vehicle rule turn supplier provenance and patching into approval and compliance gates.

Why screened out: Screened out as a stand-alone wedge in B-2.3.3: buyers need audit credibility a student team lacks, and compliance incumbents already compete. Cybersecurity stays a built-in requirement of whatever we build.

EVIDENCE FOR

- R155 is a gate to EU type approval [26]
- The BIS rule bars Chinese/Russian connected-vehicle software from MY2027 [30]

EVIDENCE AGAINST

- Audit credibility matters more than engineering skill here [46]
- Specialists such as Upstream, VicOne and PlaxidityX already compete [44]

APPENDIX · EVIDENCE AUDIT

Every material claim, audited and classified

43 market claims audited against 48 sources on 13 Oct 2026: **37** verified or corrected; **6** flagged as forecast, estimate, survey or hypothesis; **9** items still need a manual check. Full audit report. The 26 Oct company screen and the BMCs have their own source lists and were not part of this audit.

Still to confirm by hand

1. McKinsey 2035 PDF: publication month; whether L2 hits 52% in 2030 or 2035
2. NHTSA 2025 recall report PDF: exact 891 / 29.3M wording
3. QNX developer report: VP sample size behind the 52%
4. GM shareholder letters: exact ~70% margin wording; 2026 guidance reaffirmed
5. GM 10-Q: verbatim $0.5B accrual wording
6. Qualcomm and Tesla primary releases: $1.59B record wording; FSD purchase/subscription split
7. ISO/SAE 21434: 42 work-product count
8. China GB 44495 amendment dates; India AIS-189/190 status
9. Pending deals: Harman–ZF ADAS; Applied Intuition ~$30B round

Claims

Corrected in v2

WHY SDVS

$519B automotive software & electronics market by 2035; 4.5%/yr vs ~1% for vehicles

~~v1: labelled the "SDV" software & electronics market~~

To confirm: McKinsey PDF blocked to automated fetch: confirm the January 2026 publication month

[1] McKinsey Center for Future Mobility · c. Jan 2026

Verified

WHY SDVS

GM guides ~$3.1B of OnStar revenue for 2026 (+15%)

[6] General Motors · 28 Apr 2026

Corrected in v2

WHY SDVS

GM targets ~70% gross margins on OnStar and Super Cruise

~~v1: "software keeps about 70¢ of each dollar"~~

To confirm: GM shareholder letter blocked: wording read via press coverage

[8] General Motors (via Repairer Driven News) · 23 Oct 2025

Verified

WHY SDVS

Rivian's Gen 2 R1 cut ECUs from 17 to 7

[11] Rivian (Business Wire) · 6 Jun 2024

Corrected in v2

WHY SDVS

891 US vehicle recalls covered 29.3M vehicles in 2025
~~v1: “about 900 recalls, 29.2M vehicles”~~
To confirm: NHTSA PDF returned 403: figures read via a page quoting it
[2] NHTSA · Mar 2026

Forecast

WHY SDVS

OTA recall fixes could save US OEMs $1.5B by 2028
~~v1: now labelled as a 2023 forecast~~
[4] ABI Research · 3 May 2023

Corrected in v2

WHY SDVS

Chinese OEMs target 18–24-month development cycles, about half the Western norm
~~v1: “already run” 18–24-month cycles~~
[12] S&P Global Mobility AutoTechInsight · 14 Jul 2026

Verified

ECOSYSTEM

BMW buys compute directly from Qualcomm (Ride Pilot 2025; supply deal 2026)
~~v1: the Tier 1 “squeeze” is now framed as unquantified~~
[16] Reuters (via The Star) · 29 Jul 2026

Corrected in v2

ECOSYSTEM

About 30–40% of Super Cruise owners keep paying after the 3 included years
~~v1: “only ~40% renew after the trial”~~
[7] General Motors · 27 Jan 2026

Verified

ECOSYSTEM

FTC final order vs GM, 14 Jan 2026: 20-year opt-in consent; 5-year ban on sharing with consumer reporting agencies
[10] US Federal Trade Commission · 14 Jan 2026

Verified

ECOSYSTEM

VW’s Cariad lost €2.4B in 2023 (and €2.4B again in 2024)
[37] InsideEVs · 2025

Corrected in v2

ECOSYSTEM

Ford scrapped its FNV4 zonal architecture in 2025
~~v1: “scaled back” its centralized architecture~~
[38] Reuters (via Automotive News Europe) · 30 Apr 2025

Corrected in v2

ECOSYSTEM

Harman moved out of “SDV specialists” (Samsung-owned Tier 1, buying ZF ADAS); Excelfore shown instead
~~v1: Harman listed as an SDV specialist~~

[42] Brake & Front End · Dec 2025

Corrected in v2

ECOSYSTEM

"Consumers" now shows OnStar and Super Cruise as services they pay for; data partners no longer repeat Motive, and "insurance" was dropped

~~v1: GM products shown as consumers~~

[48] B-2.3.1 team · Oct 2026

Verified

VALUE CHAIN · 01

Central compute units $5B → $17B and zonal units $4B → $13B, 2030–35

[1] McKinsey Center for Future Mobility · c. Jan 2026

Verified

VALUE CHAIN · 01

Qualcomm record $1.59B automotive quarter (fiscal Q3 2026), +61% YoY

To confirm: Release body did not load: figures from consistent secondary coverage

[13] Qualcomm · Jul 2026

Estimate or survey

VALUE CHAIN · 01

30–50% compute headroom for future OTA features is one analyst's guidance

~~v1: stated as a requirement~~

[5] ABI Research (J. Hodgson) via EE Times Asia · 31 Oct 2024

Verified

VALUE CHAIN · 02

QNX in 275M+ vehicles across 45+ automakers, including the top 10

[17] BlackBerry · Apr 2026

Corrected in v2

VALUE CHAIN · 02

Android Automotive: Volvo/Polestar, GM, Renault; Ford's system is "built with Google"

~~v1: listed Ford and Renault-Nissan as AAOS users~~

[18] Ford Authority · Nov 2024

Verified

VALUE CHAIN · 02

OS & middleware ~$3B → ~$13B, 2025–35 (16.2%/yr)

[1] McKinsey Center for Future Mobility · c. Jan 2026

Corrected in v2

VALUE CHAIN · 02

RV Tech is a captive Rivian–VW joint venture, not a merchant platform vendor

~~v1: listed as a direct competitor~~

[20] Automotive World · Nov 2024

Verified

VALUE CHAIN · 03

ADAS/AD software ~$9B → ~$50B, 2025–35 (19.4%/yr; rounded endpoints give 18.7%)

[1] McKinsey Center for Future Mobility · c. Jan 2026

Verified

VALUE CHAIN · 03

Mobileye ~38M EyeQ units in 2026 outlook; Surround ADAS $100–150 ASP, ~70% gross margin

[21] Mobileye · 23 Apr 2026

Corrected in v2

VALUE CHAIN · 03

Tesla: 1.48M active FSD (Supervised) users in Q2 2026, ~45% on subscription

~~v1: "1.48M active FSD subscriptions"~~

To confirm: Purchase/subscription split from secondary sources, not Tesla's update PDF

[22] InsideEVs · Jul 2026

Verified

VALUE CHAIN · 03

L2 could be 52% of sales by 2030; L3 16% by 2035

To confirm: One write-up says 52% by 2035: check McKinsey Exhibit 1

[1] McKinsey Center for Future Mobility · c. Jan 2026

Corrected in v2

VALUE CHAIN · 04

~15% of recalled US vehicles since 2022 were eligible for an OTA remedy

~~v1: "~16% fixed over the air (Q3 2025)"~~

[3] BizzyCar · 19 Jan 2026

Corrected in v2

VALUE CHAIN · 04

Airbiquity removed from OTA players (assets bought by Karma, Feb 2024)

~~v1: Airbiquity listed as an OTA supplier~~

[39] Karma Automotive · 28 Feb 2024

Estimate or survey

VALUE CHAIN · 04

Software is ~20–25% of US recall campaigns; most are still hardware

[40] EngiMarket (analysis of NHTSA data) · 2026

Corrected in v2

VALUE CHAIN · 05

OnStar: ~12M subscribers (target 13M by end-2026), ~$20 average monthly revenue each

~~v1: "~13M subscribers paying ~$20 a month"~~

[6] General Motors · 28 Apr 2026

Verified

VALUE CHAIN · 05

Samsara $2.1B ARR, +30% YoY
[23] Samsara · 3 Sep 2026

Corrected in v2

VALUE CHAIN · 06

GM has accrued $0.5B for driver-data investigations and litigation
~~v1: "$300M"~~
To confirm: Accrual wording came from filing excerpts: confirm in the 10-Q
[9] General Motors · Jul 2026

Verified

VALUE CHAIN · 07

494 public automotive cyber incidents in 2025; 92% remote; 44% ransom-related
[25] Upstream Security · Feb 2026

Verified

VALUE CHAIN · 07

UN R155/R156 mandatory for all new EU registrations since July 2024
[26] UNECE · 2021; EU all new registrations from Jul 2024

Corrected in v2

VALUE CHAIN · 07

China (Jan 2026) and Korea (Aug 2025) enforce R155-aligned rules; India's are still draft
~~v1: "China, India and Korea have aligned drafts"~~
To confirm: Confirm China amendment dates and whether India has finalised its rules
[27] TÜV SÜD · 2025

Verified

VALUE CHAIN · 07

BIS rule: Chinese/Russian connected-vehicle software banned from MY2027, hardware from MY2030
[30] US Department of Commerce (BIS) · 14 Jan 2025

Verified

VALUE CHAIN · 07

ISO/SAE 21434 defines 42 work products (two secondary sources; the standard is paywalled)
To confirm: Standard is paywalled: confirm the count against ISO/SAE 21434
[31] ISO / SAE · 2021

Verified

VALUE CHAIN · 08

Integration, V&V services: fastest-growing slice of auto software spend, 11.4%/yr
[1] McKinsey Center for Future Mobility · c. Jan 2026

Corrected in v2

VALUE CHAIN · 08

Applied Intuition: $15B valuation (June 2025), ~$30B offers reported Oct 2026
~~v1: $15B shown as current~~
[33] Tech Startups · 13 Oct 2026

Estimate or survey

VALUE CHAIN · 08

Applied Intuition ~$830M ARR is a Sacra estimate, not a company figure

[34] Sacra · 2025

Corrected in v2

VALUE CHAIN · 08

52% of engineering VPs name integration complexity as a top challenge (QNX)

~~v1: "the top challenge", credited to McKinsey~~

To confirm: VP sample size not published in the QNX release

[36] Design News (byline by QNX) · 5 Jan 2026

Estimate or survey

VALUE CHAIN · 08

Developers cite long cycles (37%) and debugging/testing (36%); QNX 2025 vendor survey, n = 1,100

[35] QNX (BlackBerry) · 14 Oct 2025

Corrected in v2

WHITESPACE

77% of E/E architectures zonal or central by 2035, up from under 30% today

~~v1: unsourced "90%+ of OEMs committed to zonal"~~

[1] McKinsey Center for Future Mobility · c. Jan 2026

Team hypothesis

WHITESPACE

Competition ratings, open pockets and matrix placements are team hypotheses

[48] B-2.3.1 team · Oct 2026

Sources

1. **McKinsey Center for Future Mobility**, The automotive software and electronics market through 2035 (c. Jan 2026; Analyst / research firm). www.mckinsey.com/features/mckinsey-center-for-future-mobility/our-insi...
2. **NHTSA**, 2025 Annual Report: Safety Recalls (Mar 2026; Primary source). www.nhtsa.gov/sites/nhtsa.gov/files/2026-03/2025-annual-recalls-report...
3. **BizzyCar**, Quarterly recall report, Q4 2025 (OTA-eligible share, from NHTSA data) (19 Jan 2026; Analyst / research firm). bizzycar.com/blog/automotive-recall-alert-8.6-million-vehicles-recalle...
4. **ABI Research**, By 2028, automakers will save US$1.5 billion using OTA updates to fix recalled cars (3 May 2023; Analyst / research firm). abiresearch.com/press/by-2028-automakers-will-save-us15-billion-using-...
5. **ABI Research (J. Hodgson) via EE Times Asia**, Analyst commentary on compute headroom for SDV platforms (31 Oct 2024; Analyst / research firm). www.eetasia.com/support-from-chip-partners-needed-for-sdv-success/
6. **General Motors**, Q1 2026 earnings call (transcript) (28 Apr 2026; Primary source). www.fool.com/earnings/call-transcripts/2026/04/28/gm-q1-2026-earnings-...
7. **General Motors**, Q4 2025 earnings call (transcript) (27 Jan 2026; Primary source). www.fool.com/earnings/call-transcripts/2026/01/27/gm-gm-q4-2025-earnin...
8. **General Motors (via Repairer Driven News)**, Q3 2025 shareholder letter: ~70% gross margins on OnStar and Super Cruise (23 Oct 2025; Primary source). www.repairerdrivennews.com/2025/10/23/gm-reports-double-digit-revenue-...
9. **General Motors**, Form 10-Q, quarter ended 30 Jun 2026 (Jul 2026; Primary source). www.sec.gov/Archives/edgar/data/1467858/000146785826000051/gm-20260630...

10. **US Federal Trade Commission**, FTC finalizes order settling allegations GM and OnStar collected and sold geolocation data (14 Jan 2026; Primary source). www.ftc.gov/news-events/news/press-releases/2026/01/ftc-finalizes-orde...
11. **Rivian (Business Wire)**, Rivian introduces second-generation R1S and R1T (6 Jun 2024; Primary source). www.businesswire.com/news/home/20240606335706/en/Rivian-Introduces-Sec...
12. **S&P Global Mobility AutoTechInsight**, The real meaning of China speed (14 Jul 2026; Analyst / research firm). autotechinsight.spglobal.com/news/5289482/the-real-meaning-of-china-sp...
13. **Qualcomm**, Third quarter fiscal 2026 results (Jul 2026; Primary source). www.qualcomm.com/news/releases/2026/07/qualcomm-announces-third-quarte...
14. **Qualcomm**, CES 2026 release on Snapdragon automotive momentum (75M+ cockpit vehicles as of June 2025) (Jan 2026; Primary source). www.qualcomm.com/news/releases/2026/01/qualcomm-drives-the-future-of-m...
15. **Qualcomm**, Qualcomm and BMW Group unveil Snapdragon Ride Pilot (5 Sep 2025; Primary source). www.qualcomm.com/news/releases/2025/09/qualcomm-and-bmw-group-unveil-g...
16. **Reuters (via The Star)**, Report on Qualcomm's long-term compute supply deal with BMW (29 Jul 2026; Press report). www.thestar.com.my/tech/tech-news/2026/07/29/qualcomm-wins-bmw-chip-su...
17. **BlackBerry**, Form 10-K, fiscal year ended 28 Feb 2026 (QNX) (Apr 2026; Primary source). www.sec.gov/Archives/edgar/data/1070235/000107023526000039/bbry-202602...
18. **Ford Authority**, Report on Ford Digital Experience being built with Google (Nov 2024; Press report). fordauthority.com/2024/11/ford-digital-experience-developed-in-partner...
19. **Automotive World**, Report on the Eclipse S-CORE v0.5 middleware release (Nov 2025; Press report). www.automotiveworld.com/news/eclipse-s-core-releases-version-0-5-alpha...
20. **Automotive World**, Report on the launch of the Rivian–Volkswagen software joint venture (Nov 2024; Press report). www.automotiveworld.com/news/faster-leaner-more-efficient-rivian-and-v...
21. **Mobileye**, Q1 2026 earnings call (transcript) (23 Apr 2026; Primary source). www.fool.com/earnings/call-transcripts/2026/04/23/mobileye-mbly-q1-202...
22. **InsideEVs**, Report on Tesla FSD active users and purchase vs subscription split, Q2 2026 (Jul 2026; Press report). insideevs.com/news/802547/tesla-fsd-subscription-half/
23. **Samsara**, Second quarter fiscal 2027 results (3 Sep 2026; Primary source). secure.businesswire.com/news/home/20260903120692/en/Samsara-Reports-Se...
24. **Samsara**, Q4 fiscal 2026 results (5 Mar 2026; Primary source). www.samsara.com/company/news/press-releases/q4-fiscal-year-2026-result...
25. **Upstream Security**, 2026 Global Automotive & Smart Mobility Cybersecurity Report (Feb 2026; Analyst / research firm). en.prnasia.com/releases/global/ransomware-attacks-on-automotive-and-sm...
26. **UNECE**, UN Regulation No. 155 (cyber security) and No. 156 (software update) (2021; EU all new registrations from Jul 2024; Primary source). unece.org/transport/documents/2021/03/standards/un-regulation-no-155-c...
27. **TÜV SÜD**, Technical update on China's GB 44495-2024 vehicle cybersecurity standard (2025; Analyst / research firm). www.tuvsud.com/en/knowledge-hub/technical-updates/consumer-products-an...
28. **Korea Herald**, Report on Korea's vehicle cybersecurity (CSMS) certification taking effect (Aug 2025; Press report). www.koreaherald.com/article/10660606
29. **Autocar Pro**, Report on India's draft cybersecurity and software-update rules (AIS-189/190) (27 Jun 2026; Press report). www.autocarpro.in/news/india-drafts-mandatory-cybersecurity-software-u...
30. **US Department of Commerce (BIS)**, Commerce finalizes rule to secure connected vehicle supply chains (14 Jan 2025; Primary source). www.bis.gov/press-release/commerce-finalizes-rule-secure-connected-veh...
31. **ISO / SAE**, ISO/SAE 21434:2021 Road vehicles: cybersecurity engineering (work-product count via secondary sources) (2021; Primary source). www.iso.org/standard/70918.html
32. **Applied Intuition**, Series F at a $15B valuation (17 Jun 2025; Primary source). www.appliedintuition.com/news/series-f
33. **Tech Startups**, Report on investment offers to Applied Intuition at about a $30B valuation (13 Oct 2026; Press report). techstartups.com/2026/10/13/applied-intuition-receives-investment-offe...
34. **Sacra**, Company profile with Applied Intuition revenue estimate (2025; Analyst / research firm). sacra.com/c/applied-intuition/
35. **QNX (BlackBerry)**, Under the Hood: The SDV Developer Report (n = 1,100 developers) (14 Oct 2025; Vendor survey). www.nasdaq.com/press-release/qnx-research-reveals-regulatory-pressure-...
36. **Design News (byline by QNX)**, Why software-defined vehicles are the ultimate test for automakers (5 Jan 2026; Vendor survey). www.designnews.com/automotive-engineering/why-software-defined-vehicle...

37. **InsideEVs**, Report on Cariad operating losses, 2023 and 2024 (2025; Press report). insideevs.com/news/753673/vw-group-cariad-billions-losses-2024
38. **Reuters (via Automotive News Europe)**, Report on Ford cancelling its FNV4 electrical architecture (30 Apr 2025; Press report). europe.autonews.com/ford/an-ford-kills-electronics-project-0430
39. **Karma Automotive**, Karma acquires assets and key personnel from Airbiquity (28 Feb 2024; Primary source). karmaautomotive.com/news/karma-automotive-acquires-assets-and-key-pers...
40. **EngiMarket (analysis of NHTSA data)**, Analysis of NHTSA recall data, including the software share of 2025 campaigns (2026; Analyst / research firm). www.engimarket.com/blog/vehicle-recall-statistics/
41. **Frost & Sullivan**, Award release on Excelfore's OTA solutions and automaker customers (2024; Analyst / research firm). www.frost.com/news/press-releases/excelfore-recognized-with-frost-sull...
42. **Brake & Front End**, Report on Harman agreeing to acquire ZF's ADAS business (Dec 2025; Press report). www.brakeandfrontend.com/harman-to-acquire-zfs-adas-business/
43. **Synopsys**, Synopsys completes acquisition of Ansys (17 Jul 2025; Primary source). news.synopsys.com/2025-07-17-Synopsys-Completes-Acquisition-of-Ansys
44. **Elektrobit**, About Elektrobit (part of Aumovio, with PlaxidityX) (2025; Primary source). www.elektrobit.com/about/
45. **WardsAuto / NXP survey**, Article on the NXP survey of OEM SDV and zonal expectations (10 Apr 2024; Vendor survey). www.wardsauto.com/news/tackling-oems-dilemmas-in-software-defined-vehi...
46. **B-2.3.3 team draft**, Feasibility screening and engineering personas (scores are a first draft for the team to calibrate) (Oct 2026; Team assessment)
47. **B-2.3.3 team draft**, Personas and stakeholders guide: archetypes built from public research; every pain is a hypothesis (Oct 2026; Team assessment)
48. **B-2.3.1 team**, Team assessment: competition ratings, open pockets and matrix placements are hypotheses to test in SME interviews (Oct 2026; Team assessment)
49. **B-2.3.3 team report**, Feasibility Screening & Engineering Personas: Applied Intuition, Mobileye, dSPACE (every score is a hypothesis to test in Studio 1; full report kept within the team) (26 Oct 2026; Team assessment)
50. **B-2.3.2 team BMC**, Applied Intuition Business Model Canvas v2 (30 Oct 2026; Team assessment). docs/bmc/Applied_Intuition_BMC_v2.pdf
51. **B-2.3.2 team BMC**, dSPACE Business Model Canvas v2 (30 Oct 2026; Team assessment). docs/bmc/dSPACE_BMC_v2.pdf
52. **B-2.3.2 team BMC**, Mobileye Business Model Canvas v2 (30 Oct 2026; Team assessment). docs/bmc/Mobileye_BMC_v2.pdf
53. **EcoCAR Innovation Challenge**, Year 1 GM Technical Deliverable Requirements, Rev 0 (7 Aug 2026; Primary source)
54. **Growth Innovation Strategy**, DFV Wedge Hypothesis Down-Selection tool (provided at the fall workshop) (accessed 9 Nov 2026; Primary source). www.growthinnovationstrategy.com/dfv-wedge-hypothesis
55. **MarketsandMarkets via PR Newswire**, Hardware-in-the-loop market to 2030 (method undisclosed) (29 Apr 2025; Analyst / research firm). prnewswire.co.uk/news-releases/hardware-in-the-loop-market-worth-1-784...
56. **Marqstats**, Global automotive virtual ECU market, by level (2026; Analyst / research firm). marqstats.com/reports/global-automotive-virtual-ecu-market/
57. **QNX**, Developer survey of 1,100 automotive software developers (vendor-commissioned) (21 Oct 2026; Vendor survey). qnx.software/en/press-release/2026/qnx-research-finds-automotive-devel...
58. **arXiv 2601.08609**, Coverage-Guided Road Selection and Prioritization for Efficient Testing in Autonomous Driving Systems (13 Jan 2026; Primary source). arxiv.org/abs/2601.08609
59. **UNECE GRVA workshop document**, Simulation credibility framework (read as a search excerpt only) (Mar 2025; Primary source). unece.org/sites/default/files/2025-03/GRVA-WS06-03e_0.pdf

Glossary

SDV
Software-defined vehicle: features are set mainly by software on a few central and zonal computers, so they can change over the air after the sale.

ECU
Electronic Control Unit: a small computer that controls one function. Older cars have 70–100+.

Zonal architecture
A few central computers do the thinking; zonal controllers handle nearby sensors and actuators. Fewer ECUs and less wiring.

Legacy ECU migration
Moving a function from an old dedicated ECU onto zonal or central computers: splitting, porting and re-testing its software.

HIL
Hardware-in-the-Loop: a real ECU connected to a real-time computer that simulates the rest of the vehicle.

SIL
Software-in-the-Loop: testing ECU software on an ordinary computer with no real hardware.

Virtual ECU
A software model of an ECU that can run in SIL or the cloud; often not accurate enough for timing and I/O.

V&V
Verification and validation: proving software does what it should, and that it is the right thing to build.

OTA
Over-the-air update: updating vehicle software remotely over cellular instead of at a dealer.

A/B memory
Two memory banks: new software installs in the inactive one, then the ECU switches over or rolls back safely.

Bootloader
The small program on an ECU that lets new software be installed. Every supplier may use a different one.

J1939
The CAN-based network standard used in heavy trucks, buses and off-highway equipment.

AUTOSAR Classic / Adaptive
Automotive software standards: Classic for small real-time ECUs, Adaptive for powerful Linux-like computers.

DBC / ARXML
File formats describing vehicle network signals. If they disagree, integration breaks.

ISO 26262
Functional-safety standard for road-vehicle electronics; requires evidence that safety functions work.

ISO/SAE 21434
Cybersecurity engineering standard for vehicles.

UN R155 / R156
UN regulations requiring a certified cybersecurity management system (R155) and software-update management system (R156).

SBOM
Software bill of materials: a list of every software component and supplier in a product.

Homologation
Type approval: regulatory sign-off that a vehicle model meets the rules before it can be sold.

SOP
Start of production: the date a model starts mass production; everything is planned around it.

Tier 1
A supplier selling complete systems or modules directly to OEMs.

CAGR
Compound annual growth rate: the steady yearly growth that turns a start value into an end value.

ADAS

Advanced driver-assistance systems, such as adaptive cruise control and lane centering.

CAV
Connected and automated vehicle. In EcoCAR, the CAV team builds the perception and driver-assistance features.

Perception
The software that turns camera, radar and other sensor data into objects the vehicle can act on.

ML component
A piece of vehicle software whose behaviour is learned from data, such as a neural network, rather than written by hand.

MIL / PIL
Model-in-the-loop and processor-in-the-loop: testing a control model in simulation, or the compiled code on the target processor, before HIL.

Safety case
A structured argument, backed by evidence, that a system is safe enough for its intended use.

ISO/PAS 8800
Guidance on safety for AI and machine-learning components in road vehicles.

UL 4600
A safety standard for evaluating autonomous products, built around a safety case.

Wedge hypothesis
A testable guess about a narrow first market: who has a problem, what we would offer, and why they would pay.

TAM / SAM
Total addressable market (all spend on the problem) and serviceable addressable market (the part a first product could reach).

DFV
Desirability, feasibility and viability: the fall workshop's three lenses for scoring and ranking wedges.

Persona
A fictional archetype of a real job role, used to plan interviews. Not a real person.

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